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## 306.

### SUR LES CONIQUES QUI TOUCHENT DES COURBES D'ORDRE QUELCONQUE.
### EXTRAIT D'UNE LETTRE À M. CHASLES.

*[From the Comptes Rendus de l'Académie des Sciences de Paris, tom.* LIX. *(Juillet—Décembre, 1864), pp. 224–225.]*

En considérant l'expression

$$\tfrac{1}{8}(S_1 + S_2 + S_3 - 3S_4 + 3S_5)$$

que vous avez donnée (*Comptes Rendus*, t. LVIII. p. 223) pour le nombre des coniques qui touchent cinq courbes d'ordre quelconque, j'ai trouvé qu'elle peut s'écrire sous la forme que voici, savoir: en dénotant les ordres par $(m, n, p, q, r)$, et en mettant $M = m^2 - m, \ldots$, de manière que $(M, N, P, Q, R)$ seront les classes des cinq courbes, l'expression transformée est

$$(M, m)(N, n)(P, p)(Q, q)(R, r)\,[1, 2, 4, 4, 2, 1];$$

en représentant par cette notation abrégée la fonction

$$\begin{aligned} &1 \,.\, M\,N\,P\,Q\,R \\ &+ 2\Sigma\,(m\,N\,P\,Q\,R) \\ &+ 4\Sigma\,(m\,n\,P\,Q\,R) \\ &+ 4\Sigma\,(m\,n\,p\,Q\,R) \\ &+ 2\Sigma\,(m\,n\,p\,q\,R) \\ &+ 1 \,.\, m\,n\,p\,q\,r. \end{aligned}$$

En écartant les relations $M = m^2 - m, \ldots$, et en supposant seulement que $(m, n, p, q, r)$ soient les ordres, et $(M, N, P, Q, R)$ les classes des cinq courbes, la nouvelle formule s'applique aux courbes avec des points doubles ou de rebroussement; on peut même supposer que la courbe de la classe $M$ et de l'ordre $m$ se réduise à un système de $M$ points et de $m$ droites, et que les autres courbes se réduisent aussi à des systèmes de points et droites; et cela étant, on obtient une vérification immédiate de la formule. Car en choisissant dans le système qui remplace chaque courbe un élément (point ou droite) à volonté, on obtient

$$\begin{array}{rl} M\,N\,P\,Q\,R & \text{systèmes } 5p, \\ \Sigma\,m\,N\,P\,Q\,R & \text{systèmes } 4p, 1d, \\ \Sigma\,m\,n\,P\,Q\,R & \text{systèmes } 3p, 2d, \\ \Sigma\,m\,n\,p\,Q\,R & \text{systèmes } 2p, 3d, \\ \Sigma\,m\,n\,p\,q\,R & \text{systèmes } 1p, 4d, \\ m\,n\,p\,q\,r & \text{systèmes } 5d. \end{array}$$

Or la condition par rapport à la première courbe se réduit à celle de passer par l'un quelconque des $M$ points, ou de toucher l'une quelconque des $m$ droites; et de même pour les autres courbes. Donc (en entendant par le mot *toucher* appliqué à un système de points et de droites, passer par les points et toucher les droites du système) la conique doit toucher l'un quelconque des systèmes $(5p)$, ou $(4p, 1d)$, ou $(3p, 2d), \ldots$, ou $(5d)$; et pour un système de la forme

$$(5p),\quad (4p, 1d),\quad (3p, 2d),\quad (2p, 3d),\quad (1p, 4d),\quad \text{ou}\quad (5d),$$

le nombre des coniques est

$$1,\quad 2,\quad 4,\quad 4,\quad 2,\quad \text{ou}\quad 1,$$

ce qui donne pour le nombre total des coniques l'expression ci-dessus écrite.

On peut supposer que la conique, au lieu de toucher les deux courbes $m$, $n$, ait avec la seule courbe $m$ ($1^{\circ}$) un contact du deuxième ordre; ($2^{\circ}$) un contact double. Le nombre des coniques qui satisfont à l'une ou l'autre de ces conditions, et qui passent aussi par trois points donnés, a été trouvé par Steiner (*Aufgabe und Lehrsätze*, Crelle, t. XLIX. p. 273), savoir:

$$(1^{\circ})\text{ le nombre} = 3m(m-1); \qquad (2^{\circ})\text{ le nombre} = \tfrac{1}{2}(m^2 - m)(m^2 + 3m - 6);$$

j'ai vérifié d'une manière particulière ces deux résultats.

En supposant que la conique (au lieu de passer par les trois points donnés) touche les courbes $p$, $q$, $r$, je trouve pour les deux cas respectivement ces résultats,

1° le nombre $= 3(m^2 - m) \times (P,p)(Q,q)(R,r)\,[1,2,2,1]$,

2° le nombre $= \frac{1}{2}(m^2 - m) \times$
$(P,p)(Q,q)(R,r)\,[m^2 + 3m - 6,\ 2m^2 + 6m - 16,\ 4m^2 + 4m - 22,\ 4m^2 - 15]$;

formules dans lesquelles la courbe $m$ doit être une courbe sans singularité.

*Grasmere, Westmoreland, 27 Juillet, 1864.*

## 307.

### NOTE SUR LES FONCTIONS al($x$), &c., DE M. WEIERSTRASS.

*[From the Journal de Mathématiques (Liouville), tom. VII. (1862), pp. 137–142.]*

Les fonctions $\mathrm{al}(x)$, $\mathrm{al}(x)_1$, $\mathrm{al}(x)_2$, $\mathrm{al}(x)_3$ de M. Weierstrass satisfont respectivement aux équations

$$\frac{d^2\,\mathrm{al}(x)}{dx^2} + 2k'x\frac{d\,\mathrm{al}(x)}{dx} + 2kk'\frac{d\,\mathrm{al}(x)}{dk} + k^2x^2\,\mathrm{al}(x) = 0,$$
$$\frac{d^2\,\mathrm{al}(x)_1}{dx^2} + 2k'x\frac{d\,\mathrm{al}(x)_1}{dx} + 2kk'\frac{d\,\mathrm{al}(x)_1}{dk} + (k^2 + k^2x^2)\,\mathrm{al}(x)_1 = 0,$$
$$\frac{d^2\,\mathrm{al}(x)_2}{dx^2} + 2k'x\frac{d\,\mathrm{al}(x)_2}{dx} + 2kk'\frac{d\,\mathrm{al}(x)_2}{dk} + (1 + k^2x^2)\,\mathrm{al}(x)_2 = 0,$$
$$\frac{d^2\,\mathrm{al}(x)_3}{dx^2} + 2k'x\frac{d\,\mathrm{al}(x)_3}{dx} + 2kk'\frac{d\,\mathrm{al}(x)_3}{dk} + (k'^2 + k^2x^2)\,\mathrm{al}(x)_3 = 0,$$

ou, ce qui est la même chose, les fonctions

$$\mathrm{al}(x), \quad \sqrt{k}\,\mathrm{al}(x)_1, \quad \frac{\sqrt{k}}{\sqrt{k'}}\,\mathrm{al}(x)_2, \quad \frac{1}{\sqrt{k'}}\,\mathrm{al}(x)_3,$$

satisfont chacune à l'équation

$$\frac{d^2z}{dx^2} + 2k'x\frac{dz}{dx} + 2kk'\frac{dz}{dk} + k^2x^2z = 0.$$

Écrivons pour un moment $\xi$, $\kappa$ au lieu de $x$, $k$; les fonctions $\mathrm{al}(\xi)$, etc., satisfont à l'équation

$$\frac{d^2z}{d\xi^2} + 2\kappa^2\xi\frac{dz}{d\xi} + 2\kappa\kappa'^2\frac{dz}{d\kappa} + \kappa'^2\xi^2z = 0.$$

Cela étant, en posant

$$\xi = \frac{x}{\sqrt{k}}, \qquad \kappa = k,$$

on obtient

$$\frac{dz}{dx} = \frac{1}{\sqrt{k}}\frac{dz}{d\xi}, \quad \frac{d^2z}{dx^2} = \frac{1}{k}\frac{d^2z}{d\xi^2}, \quad \frac{dz}{dk} = -\frac{x}{2k\sqrt{k}}\frac{dz}{d\xi} + \frac{dz}{d\kappa},$$

et de là

$$\frac{dz}{d\xi} = \sqrt{k}\,\frac{dz}{dx}, \quad \frac{d^2z}{d\xi^2} = k\frac{d^2z}{dx^2}, \quad \frac{dz}{d\kappa} = \frac{dz}{dk} + \frac{x}{2k}\frac{dz}{dx}.$$

L'équation différentielle devient ainsi

$$k\frac{d^2z}{dx^2} + 2k'x\frac{dz}{dx} + 2kk'^2\left(\frac{dz}{dk} + \frac{x}{2k}\frac{dz}{dx}\right) + kx^2z = 0,$$

c'est-à-dire

$$\frac{d^2z}{dx^2} + \frac{1+k^2}{k}x\frac{dz}{dx} + 2k'^2\frac{dz}{dk} + x^2z = 0,$$

équation qui sera ainsi satisfaite par

$$\mathrm{al}\left(\frac{x}{\sqrt{k}}\right), \quad \sqrt{k}\,\mathrm{al}\left(\frac{x}{\sqrt{k}}\right)_1, \quad \frac{\sqrt{k}}{\sqrt{k'}}\,\mathrm{al}\left(\frac{x}{\sqrt{k}}\right)_2, \quad \frac{1}{\sqrt{k'}}\,\mathrm{al}\left(\frac{x}{\sqrt{k}}\right)_3.$$

Or en écrivant

$$k + \frac{1}{k} = \alpha,$$

l'équation en $z$ devient

$$\frac{d^2z}{dx^2} + 2\alpha x\frac{dz}{dx} - 2(\alpha^2-4)\frac{dz}{d\alpha} + x^2z = 0, \tag{1}$$

laquelle est ce que devient celle-ci

$$(1-\alpha x^2+x^4)\frac{d^2z}{dx^2} + (n-1)(\alpha x - 2x^3)\frac{dz}{dx} - 2n(\alpha^2-4)\frac{dz}{d\alpha} + n(n-1)x^2z = 0, \tag{2}$$

en y écrivant $\dfrac{x}{\sqrt{n}}$ au lieu de $x$, et puis $n = \infty$. L'équation (2), trouvée par Jacobi (*Journal de Crelle*, t. IV. p. 185, 1829), a la propriété que voici, savoir en posant

$$\alpha = k + \frac{1}{k}, \quad x = \sqrt{k}\sin\mathrm{am}\,u,$$

alors l'équation est satisfaite en prenant pour $z$ soit le numérateur, soit le dénominateur, de la fonction rationnelle de $x$ qui donne la valeur de la fonction $\sqrt{\lambda}\sin\mathrm{am}\left(\dfrac{u}{M}, \lambda\right)$, où $\lambda$, $M$ sont le module et le multiplicateur qui correspondent à la transformation de l'ordre $n$ ($n$ étant un nombre impair quelconque).

L'équation fut donnée par Jacobi sans démonstration. Je l'ai démontrée (*Camb. and Dubl. Math. Journal*, t. II. p. 256, 1847, [45]) de la manière que voici, savoir en écrivant

$$z = \left(\frac{2Kk'}{\pi}\right)^{\frac{1}{2}(m-1)}\Theta^{-n}(u)\cdot\Sigma,$$

on obtient pour $\Sigma$ l'équation

$$\frac{d^2\Sigma}{du^2} - 2nu\left(k^2 - \frac{E}{K}\right)\frac{d\Sigma}{du} + 2nkk'^2\frac{d\Sigma}{dk} = 0.$$

Cette équation, en prenant

$$\omega = \frac{u\pi K'}{K}, \quad v = \frac{u\pi u}{2K},$$

devient

$$\frac{d^2\Sigma}{dv^2} - 4\frac{d\Sigma}{d\omega} = 0,$$

équation mentionnée par Jacobi, laquelle est satisfaite par

$$\Sigma = \Theta\left(nu, \frac{nK'}{K}\right), \quad \text{ou } \Sigma = \mathrm{H}\left(nu, \frac{nK'}{K}\right);$$

cela donne pour $z$ deux valeurs qui sont le dénominateur et le numérateur de la fraction dont il s'agit. J'ose croire que ce doit être à peu près de cette manière que l'équation fut trouvée par Jacobi.

Or les solutions en question de l'équation (1) peuvent être trouvées au moyen de l'équation de Jacobi; pour cela, au lieu des valeurs ci-dessus données de $\omega$, $v$, j'écris

$$\omega = \frac{\pi K'}{K}, \quad v = \frac{\sqrt{n}\pi u}{2K},$$

ce qui conduit à la même équation

$$\frac{d^2\Sigma}{dv^2} - 4\frac{d\Sigma}{d\omega} = 0,$$

laquelle sera ainsi satisfaite par

$$\Sigma = \Theta\left(\sqrt{n}u, \frac{K'}{K}\right), \quad \Sigma = \mathrm{H}\left(\sqrt{n}u, \frac{K'}{K}\right),$$

ou, ce qui est la même chose, par

$$\Sigma = \Theta(\sqrt{n}u), \quad \Sigma = \mathrm{H}(\sqrt{n}u).$$

L'équation (2) sera donc satisfaite par

$$z = \left(\frac{2Kk'}{\pi}\right)^{\frac{1}{2}(n-1)} \Theta^{-n}(u)\,\Theta(\sqrt{n}u),$$

ou, en se souvenant que $\Theta(0) = \sqrt{\dfrac{2Kk'}{\pi}}$, par

$$z = \frac{\Theta(\sqrt{n}u)}{\Theta(0)} \cdot \frac{\Theta^n(0)}{\Theta^n(u)}.$$

Or en écrivant $\dfrac{x}{\sqrt{n}}$ au lieu de $x$, pour faire ensuite $n = \infty$, nous avons

$$\frac{x}{\sqrt{n}} = \sqrt{k}\sin\mathrm{am}\, u,$$

équation qui se réduit à

$$u = \frac{x}{\sqrt{nk}};$$

cela donne

$$\Theta(\sqrt{n}u) = \Theta\left(\frac{x}{\sqrt{k}}\right), \qquad \Theta^n u = \Theta^n(0)\, e^{n\int_0^u du\, k^2\,\mathrm{sn}^2\, u},$$

puisque

$$\Theta(u) = \Theta(0)\, e^{\int_0^u du\,(k^2\mathrm{sn}^2 u - \frac{E}{K})\cdots};$$

et $n\displaystyle\int_0^u du \int_0^u du\, k^2\sin^2\mathrm{am}\, u$, en y substituant $u = \dfrac{x}{\sqrt{nk}}$, contient le facteur $\dfrac{1}{n}$ et se réduit ainsi à zéro. Donc on obtient

$$z = \frac{\Theta\left(\dfrac{x}{\sqrt{k}}\right)}{\Theta(0)}\, e^{\frac{x^2}{2}\left(1-\frac{E}{K}\right)}$$

comme solution de l'équation (1), qui se déduit de l'équation (2) en y écrivant $\dfrac{x}{\sqrt{n}}$ au lieu de $x$ et puis $n=\infty$. Et cette valeur de $z$ est précisément la fonction $\mathrm{al}\left(\dfrac{x}{\sqrt{k}}\right)$ de M. Weierstrass. On obtient de même la solution

$$z=\frac{\mathrm{H}\left(\dfrac{x}{\sqrt{k}}\right)}{\Theta(0)}\,e^{\frac{x^2}{2}\left(1-\frac{E}{K}\right)},$$

ou, ce qui est la même chose,

$$z=\frac{\mathrm{H}\left(\dfrac{x}{\sqrt{k}}\right)}{\dfrac{1}{\sqrt{k}}\mathrm{H}'(0)}\,e^{\frac{x^2}{2}\left(1-\frac{E}{K}\right)},$$

qui est la fonction $\sqrt{k}\ \mathrm{al}\left(\dfrac{x}{\sqrt{k}}\right)_1$. Et d'une manière semblable les solutions

$$z=\frac{\Theta\left(\dfrac{x}{\sqrt{k}}+K\right)}{\Theta(0)}\,e^{\frac{x^2}{2}\left(1-\frac{E}{K}\right)},\qquad z=\frac{\Theta\left(\dfrac{x}{\sqrt{k}}+K\right)}{\Theta(0)}\,e^{\frac{x^2}{2}\left(1-\frac{E}{K}\right)},$$

qui sont les fonctions $\sqrt{\dfrac{k}{k'}}\ \mathrm{al}\left(\dfrac{x}{\sqrt{k}}\right)_2$, $\dfrac{1}{\sqrt{k'}}\mathrm{al}\left(\dfrac{x}{\sqrt{k}}\right)_3$.

J'ajoute que dans le Mémoire cité (1847) j'ai donné la suite

$$z=C_0+C_1\frac{x^2}{1\cdot2}+C_2\frac{x^4}{1\cdot2\cdot3\cdot4}+\cdots$$

où

$$\begin{aligned}
&C_0=1,\\
&C_1=0,\\
&C_2=-2,\\
&C_3=+8\alpha,\\
&C_4=-32\alpha^2-4,\\
&C_5=+128\alpha^3+96\alpha,\\
&C_6=-512\alpha^4-960\alpha^2-408,\\
&C_7=+2048\alpha^5+7168\alpha^3+7584\alpha,\\
&C_8=-8192\alpha^6-46080\alpha^4-88320\alpha^2-15384,\\
&\quad\vdots
\end{aligned}$$

de façon qu'en général

$$C_{r+2}=-(2r+1)(2r+2)\,C_r-(2r+2)\alpha C_{r+1}+2(\alpha^2-4)\frac{dC_{r+1}}{d\alpha},$$

c'est le développement de M. Weierstrass pour la fonction $\mathrm{al}\left(\dfrac{x}{\sqrt{k}}\right)$: seulement je ne connaissais pas alors l'expression finie

$$\mathrm{al}\left(\frac{x}{\sqrt{k}}\right)\ =\ \frac{1}{\Theta(0)}\,\Theta\left(\frac{x}{\sqrt{k}}\right)\,e^{\frac{x^2}{2}\left(1-\frac{E}{K}\right)}$$

de cette suite. Je remarque en passant que pour $\alpha=2$, la suite se réduit à

$$e^{\frac12x^2}=\tfrac12(e^x+e^{-x}).$$

# 308.

## ON THE △ FACED POLYACRONS, IN REFERENCE TO THE PROBLEM OF THE ENUMERATION OF POLYHEDRA.

*[From the Memoirs of the Literary and Philosophical Society of Manchester, vol. I. (1862), pp. 248–256.]*

The problem of the enumeration of polyhedra[1] is one of extreme difficulty, and I am not aware that it has been discussed elsewhere than in Mr Kirkman's valuable series of papers on this subject in the Memoirs of the Society and in the *Philosophical Transactions*. A case of the general problem is that of the enumeration of the polyhedra with trihedral summits; and Mr Kirkman in the earliest of his papers, viz. that "On the representation and enumeration of polyhedra" (*Memoirs*, vol. XII. pp. 47–70, 1854), has in fact, by an examination of the particular case, accomplished the enumeration of the octahedra with trihedral summits. A subsequent paper "On the enumeration of $x$-edra having trihedral summits and an $(x-1)$-gonal base," *Phil. Trans.* vol. XLVI. pp. 399–411, 1856), relates, as the title shows, only to a special case of the problem of the polyhedra with trihedral summits, and in this particular case the number of polyhedra is more completely determined; but the later memoirs relate to the problem in all its generality, and the above-mentioned particular problem of the enumeration of the polyhedra with trihedral summits is not, I think, anywhere resumed. Instead of the polyhedra with trihedral summits, it is really the same thing, but it is rather more convenient to consider the polyacrons with triangular faces, or as these may for shortness be called, the △ faced polyacrons; and it is intended in the present paper to give a method for the derivation of the △ faced polyacrons of a given number of summits from those of the next inferior number of summits, and to exemplify it by finding, in an orderly manner, the △ faced polyacrons up to the octacrons: thus, as regards the examples, stopping at the same point as Mr Kirkman, for although perfectly practicable it would be very tedious to carry them further, and there would be no commensurate advantage in doing so. The epithet △ faced will be omitted in the sequel, but it is to be understood throughout that I am speaking of such polyacrons only; and I shall for convenience use the epithets tripleural, tetrapleural, &c. to denote summits with three, four, &c. edges through them. The number of edges at a summit is of course equal to the number of faces, but it is the edges rather than the faces which have to be considered.

An $n$-acron has

$$n \text{ summits}, \quad 3n-6 \text{ edges}, \quad 2n-4 \text{ faces},$$

and it is easy to see that there are the following three cases only, viz.:

1. The polyacron has at least one tripleural summit.

2. The polyacron, having no tripleural summit, has at least one tetrapleural summit.

3. The polyacron, having no tripleural or tetrapleural summit, has at least twelve pentipleural summits.

In fact, if the polyacron has $c$ tripleural summits, $d$ tetrapleural summits, $e$ pentipleural summits, and so on, then we have

$$n = c + d + e + f + g + h + \&c.,$$

$$6n - 12 = 3c + 4d + 5e + 6f + 7g + 8h + \&c.,$$

[1] I use with Mr Kirkman the expression "enumeration of polyhedra" to designate the general problem, but I consider that the problem is to find the different polyhedra rather than to count them, and I consequently take the word enumeration in the popular rather than the mathematical sense.

and therefore

$$12 = 3c + 2d + e + 0f - g - 2h - \&c.,$$

or

$$3c + 2d + e = 12 + f + 2g + 3h + \&c.;$$

whence if $c = 0$ and $d = 0$, then $e = 12$ at least. It appears, moreover (since $n$ cannot be less than $e$), that any polyacron with less than 12 summits cannot belong to the third class, and must therefore belong to the first or the second class.

An $(n+1)$-acron, by a process which I call the subtraction of a summit, may be reduced to an $n$-acron; viz., the faces about any summit of the $(n+1)$-acron stand upon a polygon (not in general a plane figure) which may be called the basic polygon, and when the summit with the faces and edges belonging to it is removed, the basic polygon, if a triangle, will be a face of the $n$-acron; if not a triangle, it can be partitioned into triangles which will be faces of the $n$-acron. The annexed figures exhibit the process for the cases of a tripleural, tetrapleural and pentipleural summit respectively, which are the only cases which need be considered; these may be called the first, second and third process respectively. It is proper to remark that for the same removed summit the first process can be performed in one way only, the second process in two ways, the third in five ways; these being in fact the numbers of ways of partitioning the basic polygon.

We may in like manner, by the converse process of the addition of a summit, convert an $n$-acron into an $(n+1)$-acron; viz., it is only necessary to take on the $n$-acron a polygon of any number of sides, and make this the basic polygon of the new summit of the $(n+1)$-acron, and for this purpose to remove the faces within the polygon and substitute for them a set of triangular faces standing on the sides of the polygon and meeting in the new summit: the same figures exhibit the process for the cases of a tripleural, tetrapleural and pentipleural summit respectively, which (as for the subtractions) are the only cases which need be considered. It may be noticed that for the same basic polygon the process is in each case a unique one; the process is said to be the first, second, or third process, according as the new summit is tripleural, tetrapleural, or pentipleural.

*[The original page exhibits small woodcut figures illustrating the subtraction/addition processes for tripleural, tetrapleural, and pentipleural summits; these are omitted here.]*

Now, reverting to the before-mentioned division of the polyacrons into three classes, an $(n+1)$-acron of the first class may by the first process of subtraction be reduced to an $n$-acron, and conversely it can be by the first process of addition derived from an $n$-acron. An $(n+1)$-acron of the second class, as having a tetrapleural summit, may by the second process of subtraction be reduced to an $n$-acron, and conversely it can be by the second process of addition derived from an $n$-acron. And in like manner, an $(n+1)$-acron of the third class, as having a pentipleural summit, may be by the third process of subtraction reduced to an $n$-acron, and conversely it may be by the third process of addition derived from an $n$-acron.

Hence all the $(n+1)$-acrons can be by the first, second and third processes of addition respectively derived from the $n$-acrons. It is to be observed that all the $(n+1)$-acrons of the first class are obtained by the first process; the second process is only required for finding the $(n+1)$-acrons of the second class; and these being all obtained by means of it, the third process is only required for finding the $(n+1)$-acrons of the third class. Hence the second process need only be made use of when the $n$-acron has no tripleural summit, or when it has only one tripleural summit, or when, having two tripleural summits, they are the opposite summits of two adjacent faces. In the last-mentioned two cases respectively it is only necessary to consider the basic quadrangles which pass through the single tripleural summit and the basic quadrangle which passes through the two tripleural summits; for with any other basic quadrangle the derived $(n+1)$-acron would retain a tripleural summit, and would consequently be of the first class. The condition is more simply expressed as follows, viz.: The second process need only be employed when there is on the $n$-acron a basic quadrangle the summits of which are at least of the number of edges shown in the annexed figure, and all the other summits are at least 4-pleural. Again, by the third process (as already mentioned) we seek only to obtain the $(n+1)$-acrons of the third class; the process need only be applied to the $n$-acrons for which there exists a basic pentagon the summits of which are at least of the number of edges shown in the annexed figure, all the other summits being at least 5-pleural; for it is only in this case that the derived $(n+1)$-acron will be of the third class. The condition just referred to obviously implies that the $n$-acron is of the second or third class. It is to be noticed that in applying the foregoing principles to the formation of the polyacrons as far as the 11-acrons we are only concerned with the first and second processes.

*[The original includes small diagrams of a basic quadrangle (with corner degrees $4,3,4,3$) and a basic pentagon (with corner degrees $4,4,5,4,4$) illustrating the conditions just stated; these woodcuts are omitted here.]*

Consider the entire series of $n$-acrons, say $A$, $B$, $C$, &c., and suppose that the $n$-acron $A$ gives rise to a certain number, say $P$, $Q$, $R$, $S$ of $(n+1)$-acrons, the $(n+1)$-acron $P$ is of course derivable from the $n$-acron $A$, but it may be derivable from other $n$-acrons, suppose from the $n$-acrons $B$ and $C$. Then in considering the $(n+1)$-acrons derived from $B$, one of these will of course be found to be the $(n+1)$-acron $P$, and it is only the remaining $(n+1)$-acrons derived from $B$ which are or may be $(n+1)$-acrons not already previously obtained as $(n+1)$-acrons derived from $A$. And if in this manner, as soon as each $(n+1)$-acron is obtained, we apply to it the process of subtraction so as to ascertain the entire series of $n$-acrons from which it is derivable, and, in forming the $(n+1)$-acrons derived from these, take account of the $(n+1)$-acrons already previously obtained and found to be derivable from these, we should obtain without any repetitions the entire series of the $(n+1)$-acrons.

For merely finding the number of the $(n+1)$-acrons, a more simple process might be adopted: say that an $n$-acron is $p$-wise generating when it gives rise to a number $p$ of $(n+1)$-acrons, and that it is $q$-wise generable when it can be derived from a number $q$ of $(n+1)$-acrons; and assume that a given $n$-acron is $(y_1 + \frac{1}{2}y_2 + \frac{1}{3}y_3 + \text{\&c.})$-wise generating, viz. that it gives rise to a number $y_1$ of $(n+1)$-acrons which are 1-wise generable, a number $y_2$ of $(n+1)$-acrons which are 2-wise generable, and so on; these forming the sum

$$\Sigma\,(y_1 + \tfrac{1}{2}y_2 + \tfrac{1}{3}y_3 + \cdots),$$

where $\Sigma$ refers to the entire series of the $n$-acrons, it is clear that every $m$-wise generable $(n+1)$-acron will in respect of each of the $n$-acrons from which it is derivable be reckoned as $\dfrac{1}{m}$, that is, it will be in the entire sum reckoned as 1, and the sum in question will consequently be the number of the $(n+1)$-acrons.

The figures of the polyacrons comprised in the annexed Tables show the application of the method to the genesis of the polyacrons as far as the octacrons, in which the numbers indicate the nature of the different summits, according to the number of edges through each summit, viz., 3 a tripleural summit, 4 a tetrapleural summit, and so on. It will be noticed that there is only a single case in which this notation is insufficient to distinguish the polyacron, viz., among the octacrons there are two forms each of them with the same symbol 33445566; the inspection of the figures shows at once that these are wholly distinct forms, for in the first of them, viz. that derived from 3344555, each of the tripleural summits stands upon a basic triangle 456, while in the other of them, that from 3444555, each of the tripleural summits stands upon a basic triangle 566. But the symbol is merely generic, and of course in the polyacrons of a greater number of summits it may very well happen that a considerable number of polyacrons are comprised in the same genus.

The following remarks on the derivation of the octacrons from the heptacrons will further illustrate the method:

1. The heptacron 3335556 has three kinds of faces,[2] viz. 356, 355 [555], the first process consequently gives rise to 3 octacrons. As the heptacron has more than two tripleural summits the second process is not applicable.

2. The heptacron 3344466 has three kinds of faces, viz.: 366, 346 and 446, and the first process gives therefore 3 octacrons. The heptacron has only two tripleural summits, and they are disposed in the proper manner; the second process gives therefore 1 octacron.

3. The heptacron 3344556 has five kinds of faces, viz. 345, 346, 356, 456 and 455, and the first process consequently gives 5 octacrons. The heptacron has two tripleural summits, but they are not disposed in such manner as to render the second process applicable.

4. The heptacron 3444555 has four kinds of faces, viz. 355, 455, 445 and 444, and the first process gives therefore 4 octacrons. The heptacron has one tripleural summit, and the basic quadrangles 3545 which belong to it are of the same kind; the second process gives therefore 1 octacron.

5. The heptacron 4444455 has only one kind of face, viz. 445, and the first process gives therefore 1 octacron. There are two kinds of basic quadrangles, viz. 4545 and 4445, and the second process gives therefore 2 octacrons.

The number of octacrons would thus be 20, but by passing back from the octacrons to the heptacrons, it is found that there are in fact only 14 octacrons. Thus the octacron 33336666 has only one kind of tripleural summit 666 (the summit is here indicated by the symbol of the basic polygon) and the octacron is thus seen to be derivable from a single heptacron only, viz. the heptacron 3335556 from which it was in fact derived. But the octacron 33345567 has three kinds of tripleural summits, viz. 567, 557 and 467, and it is consequently derivable from three heptacrons, viz. the heptacrons 3335556, 3344466 and 3344555, and so on. The passage to the heptacrons from an octacron with one or more tripleural summits is of course always by the first process, but for the last two octacrons, which have no tripleural summits, the passage back to the heptacrons is by the second process: thus for the octacron 44445555 we have but one kind of tetrapleural summit 4555; but as opposite pairs of summits of the basic quadrangle are of different kinds, viz. 45 and 55, we obtain two heptacrons, viz. 3444555 and 4444455. The octacron 44444466 has but one kind of tetrapleural summit, viz. 4646, and the pairs of opposite summits of the basic quadrangle being of the same kind 46, we obtain from it only the heptacron 4444455.

It may be remarked that for the five heptacrons respectively the values of the sum $y_1 + \frac{1}{2}y_2 + \frac{1}{3}y_3 + \cdots$ are

$$1+\tfrac{1}{2}+\tfrac{1}{3},\quad \tfrac{1}{2}+1+\tfrac{1}{2}+\tfrac{1}{2},\quad \tfrac{1}{2}+\tfrac{1}{2}+\tfrac{1}{2}+\tfrac{1}{2}+\tfrac{1}{2},\quad 1+\tfrac{1}{2}+\tfrac{1}{2}+\tfrac{1}{2}+\tfrac{1}{2},\quad 1+\tfrac{1}{2}+\tfrac{1}{2},$$

giving for $\Sigma\,(y_1 + \frac{1}{2}y_2 + \frac{1}{3}y_3 + \cdots)$ the value 14, as it should do.

[Plate — see scan: I (facing p. 44 in the original) presents the family-tree diagram of the △-faced polyacrons from 3333 (the tetrahedron) through the octacrons, together with line-drawings of selected polyacrons (the hexacron 33444, the heptacrons 334455, 334445, 444444, etc., and the octacrons 33336666, 33445566, 33455556, 33555555, 34444566, 34445556, 44445555, 44444466). The plate is omitted here; consult the printed source for the figure.]

[2] It is hardly necessary to remark that it must not be imagined that in general all the faces denoted by a symbol such as 355 (which determines only the nature of the summits on the face) are faces of the same kind, but this is so in the cases referred to in the text.

## 309.

### NOTE ON THE THEORY OF DETERMINANTS.

*[From the Philosophical Magazine, vol. XXI. (1861), pp. 180–185.]*

The following mode of arrangement of the developed expression of a determinant had presented itself to me as a convenient one for the calculation of a rather complicated determinant of the fifth order; but I have since found that it is in effect given, although in a much less compendious form, in a paper by J. N. Stockwell, "On the Resolution of a System of Symmetrical Equations with Indeterminate Coefficients," *Gould's Ast. Journal*, No. 139 (Cambridge, U. S., Sept. 10, 1860).

Suppose that the determinant

$$\begin{vmatrix} 11 & 12 & 13 \\ 21 & 22 & 23 \\ 31 & 32 & 33 \end{vmatrix}$$

is represented by $\{123\}$, and so for a determinant of any order $\{123\ldots n\}$.

Let $|1|$, $|2|$, $|12|$, $|123|$, &c., denote as follows: viz.

$$\begin{aligned} |1| &= 11, \quad |2| = 22, \ \&\text{c.} \\ |12| &= 12\cdot 21, \\ &\ \vdots \\ |123| &= 12\cdot 23\cdot 31, \end{aligned}$$

where it is to be noticed that, with the same two symbols, e.g. 1 and 2, there is but one distinct expression $|12|$ (in fact $|21| = 21\cdot 12 = |12|$); with the same three symbols, $1, 2, 3$, there are two distinct expressions, $|123|$ ($= 12\cdot 23\cdot 31$) and $|132|$ ($= 13\cdot 32\cdot 21$); and generally with the same $m$ symbols $1, 2, 3, \ldots, m$, there are $1\cdot 2\cdot 3\cdots(m-1)$ distinct expressions $|123\ldots m|$, which are obtained by permuting in every possible manner all but one of the $m$ symbols.

This being so, and writing for greater simplicity $|1|2|$ to denote the product $|1|\times|2|$, and so in general, the values of the determinants $\{12\}$, $\{123\}$, $\{1234\}$, $\{12345\}$, &c. are as follows: viz.

$$\begin{aligned} \{12\} &= +|1|2| \ - \ |12|, \\ \{123\} &= +|1|2|3| \ - \ |12|3| \ - \ |2|13| \ - \ |3|12| \ + \ |123| \ + \ |132|, \\ \{1234\} &= +|1|2|3|4| \ - \ |12|3|4| \text{ (6 terms)} \ + \ |123|4| \text{ (8 terms)} \\ &\quad + |12|34| \text{ (3 terms)} \ - \ |1234| \text{ (6 terms)}, \\ \{12345\} &= +|1|2|3|4|5| \ - \ |12|3|4|5| \ (10) \ + \ |123|4|5| \ (20) \\ &\quad + |12|34|5| \ (15) \ - \ |1234|5| \ (30) \ - \ |123|45| \ (20) \ + \ |12345| \ (24), \end{aligned}$$

with term-counts $1{+}1 = 2$, $1{+}3{+}3{-} = 6$, $1{+}6{+}8{+}3{+}6 = 24$, and $1{+}10{+}20{+}15{+}30{+}20{+}24 = 120$ respectively.

*[The original page sets these out as a tabular display with running counts* $60 + 60 = 120$ *for* $\{12345\}$*; we give the algebraic content above.]*

where, as regards the signs, it is to be observed that there is a sign $-$ for each compartment $|\ |$ containing an even number of symbols; thus in the expression for $\{1234\}$, the terms $|1|2|3|4|$ have the sign $-\cdot- = +$, and the terms $|1234|$ the sign $-$. Or, what comes to the same thing; when $n$ is even, the sign is $+$ or $-$ according as the number of compartments is even or odd; and contrariwise when $n$ is odd. As regards the remaining part of the expression, this merely exhibits the partitions of a set of $n$ things; and the formula for the several determinants up to the determinant of a given order are all of them obtained by means of the form

[the partition diagram, carried up to order 7 in the original]

which is carried up to the order 7, but which can be further extended without any difficulty whatever.

It is perhaps hardly necessary, but I give at full length the expressions of the determinant of the third order: this is

$$\begin{aligned}\{123\} = &+|1|2|3| \\ &-|12|3| \\ &-|2|13| \\ &-|3|12| \\ &+|123| \\ &+|132|,\end{aligned}$$

and by writing down in like manner the expression for the twenty-four terms of the determinant of the fourth order, the notation will become perfectly clear.

The formula hardly requires a demonstration. The terms of a determinant $\{123\ldots m\}$, for example the determinant $\{1234\}$, are obtained by permuting in every possible manner the symbols in either column, say the second column, of the arrangement

$$\begin{matrix}1 & 1\\ 2 & 2\\ 3 & 3\\ 4 & 4\end{matrix}$$

and prefixing the sign (+ or −) of the arrangement; and the resulting arrangements, for instance

$$+\begin{matrix}1 & 1\\ 2 & 2\\ 3 & 3\\ 4 & 4\end{matrix}, \qquad -\begin{matrix}1 & 2\\ 2 & 1\\ 3 & 3\\ 4 & 4\end{matrix}, \qquad -\begin{matrix}1 & 2\\ 2 & 3\\ 3 & 4\\ 4 & 1\end{matrix},$$

are interpreted either into $+11\cdot 22\cdot 33\cdot 44$, $-12\cdot 21\cdot 33\cdot 44$, $-12\cdot 23\cdot 34\cdot 41$, or in the notation of the formula, into

$$+|1|2|3|4|, \qquad -|12|3|4|, \qquad -|1234|;$$

and so in general.

Suppose that any partition of $n$ contains $\alpha$ compartments each of $a$ symbols, $\beta$ compartments each of $b$ symbols... ($a, b, \ldots$ being all of them different and greater than unity), and $p$ compartments each of a single symbol, we have

$$n = \alpha a + \beta b + \cdots + p;$$

and writing, as usual, $\Pi a = 1\cdot 2\cdot 3 \cdots a$, &c., the number of ways in which the symbols $1, 2, 3, \ldots, n$ can be so arranged in compartments is

$$\frac{\Pi n}{(\Pi a)^\alpha (\Pi b)^\beta \cdots \Pi\alpha\, \Pi\beta\, \cdots \Pi p},$$

but each such arrangement gives $(\Pi(a-1))^\alpha \cdot (\Pi(b-1))^\beta\, \cdots$ terms of the determinant, and the corresponding number of terms therefore is

$$\frac{\Pi n}{a^\alpha b^\beta \cdots \Pi\alpha\, \Pi\beta\, \cdots \Pi p}.$$

The whole number of terms of the determinant is $\Pi n$, and we have thus the theorem

$$1 = \Sigma \frac{1}{a^\alpha b^\beta \cdots \Pi\alpha\, \Pi\beta\, \cdots \Pi p},$$

in which the summation corresponds to all the different partitions $n = \alpha a + \beta b + \cdots + p$, where $a, b, \ldots$ are all of them different and greater than unity; a theorem given in Cauchy's *Mémoire sur les Arrangements* &c., 1844. But it is to be noticed also that, the number of the positive and negative terms being equal, we have besides

$$0 = \Sigma \frac{(-)^{n-\alpha-\beta-\cdots-p}}{a^\alpha b^\beta \cdots \Pi\alpha\, \Pi\beta\, \cdots \Pi p},$$

or, what is the same thing,

$$0 = \Sigma \frac{(-)^{\alpha+\beta+\cdots}}{a^\alpha b^\beta \cdots \Pi\alpha\, \Pi\beta\, \cdots \Pi p};$$

and thence also

$$\tfrac{1}{2} = \Sigma \frac{1}{a^\alpha b^\beta \cdots \Pi\alpha\, \Pi\beta\, \cdots \Pi p},$$

where, as before, $n = \alpha a + \beta b + \cdots + p$ ($a, b, \ldots$ being all different and greater than unity); but the summation is restricted either to the partitions for which $n - \alpha - \beta - \cdots - p$ is even, or else to those for which $n - \alpha - \beta - \cdots - p$ is odd.

The formula affords a proof of the fundamental property of skew symmetrical determinants. In such a determinant we have not only $12 = -21$, &c., but also $11 = 0$, &c. Suppose that $n$, the order of the determinant, is odd; then in each line of the expression

$$\{123\ldots n\} = |1|2|\ldots|n| \ \pm \&\text{c}.$$

of the determinant, there is at least one compartment $|1|$ or $|123|$ &c. containing an odd number of symbols: let $|123|$ be such a compartment, then the determinant contains the terms $|123|P$ and $|132|P$ (where $P$ represents the remaining compartments), that is, $12 \cdot 23 \cdot 31 \cdot P$ and $13 \cdot 32 \cdot 21 \cdot P$. But in virtue of the relations $12 = -21$, &c., we have

$$12 \cdot 23 \cdot 31 = -13 \cdot 32 \cdot 21;$$

and so in all similar cases; that is, the terms destroy each other, or the skew symmetrical determinant of an odd order is equal to zero.

The like considerations show that a skew symmetrical determinant of an even order is a perfect square. In fact, considering for greater simplicity the case $n = 4$, any line in the foregoing expression of $\{1234\}$ for which a compartment contains an odd number of symbols, gives rise to terms which destroy each other, and may be omitted. The expression thus reduces itself to

$$\begin{aligned} \{1234\} = {} & + |12|34| \quad 3 \text{ terms} \\ & - |1234| \quad 6 \text{ terms}, \end{aligned}$$

which is in fact the square of

$$12 \cdot 34 + 13 \cdot 42 + 14 \cdot 23;$$

for the square of a term, say $12 \cdot 34$, is $12^2 \cdot 34^2$ or $12 \cdot 21 \cdot 34 \cdot 43$, that is, $|12|34|$, and the double of the product of two terms, say $12 \cdot 34$ and $13 \cdot 42$, is $2 \cdot 12 \cdot 34 \cdot 13 \cdot 42$, or $-12 \cdot 24 \cdot 43 \cdot 31 - 13 \cdot 34 \cdot 42 \cdot 21$, that is $-|1243| - |1342|$, and so for the other similar terms, and we have

$$\{1234\} = (12 \cdot 34 + 13 \cdot 42 + 14 \cdot 23)^2;$$

and so in general, $n$ being any even number, the skew symmetrical determinant $\{123\ldots n\}$ is equal to the square of the Pfaffian $12\ldots n$, where the law of these Pfaffian functions is

$$\overline{12\cdot 34} = 12\cdot 34 + 13\cdot 42 + 14\cdot 23,$$
$$\overline{1234\cdot 56} = 12\cdot 3456 + 13\cdot 4562 + 14\cdot 5623 + 15\cdot 6234 + 16\cdot 2345,$$

where, in the second equation, 3456, &c. are Pfaffians, viz. $3456 = 34\cdot 56 + 35\cdot 64 + 36\cdot 45$; and so on.

*2, Stone Buildings, W.C., December 28, 1860.*

## 310.

### NOTE ON MR JERRARD'S RESEARCHES ON THE EQUATION OF THE FIFTH ORDER.

*[From the Philosophical Magazine, vol.* XXI. *(1861), pp. 210–214.]*

Functions of the same set of quantities which are, by any substitution whatever, simultaneously altered or simultaneously unaltered, may be called *homotypical*[3]. Thus all symmetric functions of the same set of quantities are homotypical: $(x + y - z - w)^2$ and $xy + zw$ are homotypical, &c.

It is one of the most beautiful of Lagrange's discoveries in the theory of equations, that, given the value of any function of the roots, the value of any homotypical function may be rationally determined; in other words, that any homotypical function whatever is a rational function of the coefficients of the equation and of the given function of the roots.

The researches of Mr Jerrard are contained in his work, *An Essay on the Resolution of Equations*, London, Taylor and Francis, 1859. The solution of an equation of the fifth order is made to depend on an equation of the sixth order in $W$; and he conceives that he has shown that one of the roots of this equation is a rational function of another root: "The equation for $W$ will therefore belong to a class of equations of the sixth degree, the resolution of which can, as Abel has shown, be effected by means of equations of the second and third degrees; whence I infer the possibility of solving any proposed equation of the fifth degree by a finite combination of radicals and rational functions."

The above property of rational expressibility, if true for $W$, will be true for any function homotypical with $W$; and conversely. I proceed to inquire into the form of the function $W$.

The function $W$ is derived from the function $P$, which denotes any one of the quantities $p_1$, $p_2$, $p_3$. And if $x_1$, $x_2$, $x_3$, $x_4$, $x_5$ are the roots of the given equation of the fifth order, and if $\alpha, \beta, \gamma, \delta, \epsilon$ represent in an undetermined or arbitrary order of succession the five indices $1, 2, 3, 4, 5$, and if $\iota$ denote an imaginary fifth root of unity (I conform myself to Mr Jerrard's notation), then $p_1, p_2, p_3$, and the other auxiliary quantities $t$, $u$, are obtained from the system

[3]The *à priori* demonstration shows the cases of failure. Suppose that the roots of a biquadratic equation are $1, 3, 5, 9$; then, given $a + b = 8$, we know that either $a = 3$, $b = 5$, or else $a = 5$, $b = 3$, and in either case $ab = 15$; hence in the present case (which represents the general case), $a + b$ being known, the homotypical function $ab$ is rationally determined. But if the roots are $1, 3, 5, 7$ (where $1 + 7 = 3 + 5$), then, given $a + b = 8$, this is satisfied by $(a, b = 3, 5)$ or by $(a, b = 1, 7)$, and the conclusion is $ab = 15$ or $7$; so that here $ab$ is determined, not as before, rationally, but by a quadratic equation.

of equations:

$$\begin{aligned}
x_\alpha^4 + p_1 x_\alpha^3 + p_2 x_\alpha^2 + p_3 &= t + u,\\
x_\beta^4 + p_1 x_\beta^3 + p_2 x_\beta^2 + p_3 &= \iota t + \iota^4 u,\\
x_\gamma^4 + p_1 x_\gamma^3 + p_2 x_\gamma^2 + p_3 &= \iota^2 t + \iota^3 u,\\
x_\delta^4 + p_1 x_\delta^3 + p_2 x_\delta^2 + p_3 &= \iota^3 t + \iota^2 u,\\
x_\epsilon^4 + p_1 x_\epsilon^3 + p_2 x_\epsilon^2 + p_3 &= \iota^4 t + \iota\, u.
\end{aligned}$$

If from these equations we seek for the values of $p_1, p_2, p_3, t, u$, we have

$$1 : p_1 : p_2 : p_3 : -t : -u = \Pi_1 : \Pi_2 : \Pi_3 : \Pi_4 : \Pi_5 : \Pi_6,$$

where $\Pi_1, \Pi_2, \ldots$ denote the determinants formed out of the matrix

$$\begin{pmatrix}
x_\alpha^4 & x_\alpha^3 & x_\alpha^2 & 1 & 1 & 1\\
x_\beta^4 & x_\beta^3 & x_\beta^2 & 1 & \iota & \iota^4\\
x_\gamma^4 & x_\gamma^3 & x_\gamma^2 & 1 & \iota^2 & \iota^3\\
x_\delta^4 & x_\delta^3 & x_\delta^2 & 1 & \iota^3 & \iota^2\\
x_\epsilon^4 & x_\epsilon^3 & x_\epsilon^2 & 1 & \iota^4 & \iota
\end{pmatrix},$$

i.e., denoting the columns of this matrix by $1, 2, 3, 4, 5, 6$, we have $\Pi_1 = 23456$, $\Pi_2 = -34561$, $\Pi_3 = 45612$, &c. In particular, the value of $\Pi_1$ is

$$\begin{vmatrix}
x_\alpha^3 & x_\alpha^2 & 1 & 1 & 1\\
x_\beta^3 & x_\beta^2 & 1 & \iota & \iota^4\\
x_\gamma^3 & x_\gamma^2 & 1 & \iota^2 & \iota^3\\
x_\delta^3 & x_\delta^2 & 1 & \iota^3 & \iota^2\\
x_\epsilon^3 & x_\epsilon^2 & 1 & \iota^4 & \iota
\end{vmatrix}$$

and developing, and putting for shortness $\{\alpha\beta\} = x_\alpha x_\beta (x_\alpha - x_\beta)$, &c., we have

$$\begin{aligned}
\Pi_1 = (\{\alpha\beta\} + \{\beta\gamma\} + \{\gamma\delta\} + \{\delta\epsilon\} + \{\epsilon\alpha\})\,(-2 + \iota^2 - \iota^3)&\\
+ (\{\alpha\gamma\} + \{\gamma\epsilon\} + \{\epsilon\beta\} + \{\beta\delta\} + \{\delta\alpha\})\,(+\iota + 2\iota^2 - 2\iota^3 - 2\iota^4)&;
\end{aligned}$$

and this is also the form of the other determinants, the only difference being as to the meaning of the symbol $\{\alpha\beta\}$, which, however, in each case denotes a function such that $\{\alpha\beta\} = -\{\beta\alpha\}$. Writing for greater shortness,

$$\{\alpha\beta\gamma\delta\epsilon\} = \{\alpha\beta\} + \{\beta\gamma\} + \{\gamma\delta\} + \{\delta\epsilon\} + \{\epsilon\alpha\},$$

and in like manner

$$\{\alpha\gamma\epsilon\beta\delta\} = \{\alpha\gamma\} + \{\gamma\epsilon\} + \{\epsilon\beta\} + \{\beta\delta\} + \{\delta\alpha\},$$

$\Pi_1$ is an unsymmetric linear function (without constant term) of $\{\alpha\beta\gamma\delta\epsilon\}$, $\{\alpha\gamma\epsilon\beta\delta\}$; or, what is all that is material, it is an unsymmetric function, containing only odd powers, of $\{\alpha\beta\gamma\delta\epsilon\}$, $\{\alpha\gamma\epsilon\beta\delta\}$.

If for

$$\alpha\ \ \beta\ \ \gamma\ \ \delta\ \ \epsilon$$

we substitute any one of the five arrangements

$$\begin{matrix}
\alpha & \beta & \gamma & \delta & \epsilon,\\
\beta & \gamma & \delta & \epsilon & \alpha,\\
\gamma & \delta & \epsilon & \alpha & \beta,\\
\delta & \epsilon & \alpha & \beta & \gamma,\\
\epsilon & \alpha & \beta & \gamma & \delta,
\end{matrix}$$

then $\{\alpha\beta\gamma\delta\epsilon\}$ and $\{\alpha\gamma\epsilon\beta\delta\}$ will in each case remain unaltered.

But if we substitute any one of the five arrangements

$$\begin{array}{ccccc} \alpha & \epsilon & \delta & \gamma & \beta, \\ \epsilon & \delta & \gamma & \beta & \alpha, \\ \delta & \gamma & \beta & \alpha & \epsilon, \\ \gamma & \beta & \alpha & \epsilon & \delta, \\ \beta & \alpha & \epsilon & \delta & \gamma, \end{array}$$

then in each case $\{\alpha\beta\gamma\delta\epsilon\}$ and $\{\alpha\gamma\epsilon\beta\delta\}$ will be changed into $-\{\alpha\beta\gamma\delta\epsilon\}$ and $-\{\alpha\gamma\epsilon\beta\delta\}$ respectively. Hence $\Pi_1$ remains unaltered by any one of the first five substitutions; and it is changed into $-\Pi_1$ by any one of the second five substitutions. And the like being the case as regards $\Pi_2$, &c., it follows that the quotient $\Pi_1 \div \Pi_2$, or say $P$, remains unaltered by any one of the ten substitutions. Now the 120 permutations of $\alpha, \beta, \gamma, \delta, \epsilon$ can be obtained as follows, viz. by forming the 12 different pentagons which can be formed with $\alpha, \beta, \gamma, \delta, \epsilon$ (treated as five points), and reading each of them off in either direction from any angle. To each of the 12 pentagons there corresponds a distinct value of $P$, but such value is not altered by the different modes of reading off the pentagon; $P$ is consequently a 12-valued function.

But there is a more simple form of the analytical expression of such a 12-valued function; in fact, if $[\alpha\beta]$ be any function which is not altered by any one of the above ten substitutions—if, for instance, $[\alpha\beta]$ is a symmetrical function of $x_\alpha$, $x_\beta$, and

$$[\alpha\beta\gamma\delta\epsilon] = [\alpha\beta] + [\beta\gamma] + [\gamma\delta] + [\delta\epsilon] + [\epsilon\alpha],$$

and therefore

$$[\alpha\gamma\epsilon\beta\delta] = [\alpha\gamma] + [\gamma\epsilon] + [\epsilon\beta] + [\beta\delta] + [\delta\alpha],$$

then any unsymmetrical function of $[\alpha\beta\gamma\delta\epsilon]$ and $[\alpha\gamma\epsilon\beta\delta]$ will be a 12-valued function homotypical with $P$.

Mr Jerrard's function $W$ is the sum of two values of his function $P$; the substitution by which the second is derived from the first can only be that which interchanges the two functions $[\alpha\beta\gamma\delta\epsilon]$ and $[\alpha\gamma\epsilon\beta\delta]$; and hence any symmetrical function of $[\alpha\beta\gamma\delta\epsilon]$ and $[\alpha\gamma\epsilon\beta\delta]$ is a function homotypical with Mr Jerrard's $W$; such symmetric function is in fact a 6-valued function only. Indeed it is easy to see that the twelve pentagons correspond together in pairs, either pentagon of a pair being derived from the other one by stellation, and the six values of the function in question corresponding to the six pairs of pentagons respectively.

Writing with Mr Cockle and Mr Harley,

$$\tau = x_\alpha x_\beta + x_\beta x_\gamma + x_\gamma x_\delta + x_\delta x_\epsilon + x_\epsilon x_\alpha,$$
$$\tau' = x_\alpha x_\gamma + x_\gamma x_\epsilon + x_\epsilon x_\beta + x_\beta x_\delta + x_\delta x_\alpha,$$

then $(\tau + \tau'$ is a symmetrical function of all the roots, and it must be excluded; but) $(\tau - \tau')^2$ or $\tau\tau'$ are each of them 6-valued functions of the form in question, and either of these functions is linearly connected with the Resolvent Product. In Lagrange's general theory of the solution of equations, if

$$ft = x_1 + \iota x_2 + \iota^2 x_3 + \iota^3 x_4 + \iota^4 x_5,$$

then the coefficients of the equation the roots whereof are $(ft)^5$, $(ft^2)^5$, $(ft^3)^5$, $(ft^4)^5$, and in particular the last coefficient $(ft \cdot ft^2 \cdot ft^3 \cdot ft^4)^5$ are determined by an equation of the sixth degree; and this last coefficient is a perfect fifth power, and its fifth root, or $ft \cdot ft^2 \cdot ft^3 \cdot ft^4$, is the function just referred to as the Resolvent Product.

The conclusion from the foregoing remarks is that if the equation for $W$ has the above property of the rational expressibility of its roots, the equation of the sixth order resulting from Lagrange's general theory has the same property.

I take the opportunity of adding a simple remark on cubic equations. The principle which furnishes what in a foregoing foot-note is called the *à priori* demonstration of Lagrange's theorem is that an equation need never contain extraneous roots; a quantity which has only one value will, if the investigation is properly conducted, be determined in the first instance by a linear equation; one which has two values by a quadratic equation, and so on; there is always enough, and not more than enough, to determine what is required.

*[Paper 310 continues on p. 54 of Vol. V (beyond the present excerpt).]*

## Original scan reference pages

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# 306.

## SUR LES CONIQUES QUI TOUCHENT DES COURBES D'ORDRE QUELCONQUE. EXTRAIT D'UNE LETTRE À M. CHASLES.

[From the *Comptes Rendus de l'Académie des Sciences de Paris*, tom. LIX. (*Juillet—Décembre*, 1864), pp. 224—225.]

En considérant l'expression

$$S_5\,(S_5 + S_4 + S_3 - 3S_2 + 3S_1)$$

que vous avez donnée (*Comptes Rendus*, t. LVIII. p. 223) pour le nombre des coniques qui touchent cinq courbes d'ordre quelconque, j'ai trouvé qu'elle peut s'écrire sous la forme que voici, savoir: en dénotant les ordres par $(m, n, p, q, r)$, et en mettant $M = m^2 - m, \ldots$, de manière que $(M, N, P, Q, R)$ seront les classes des cinq courbes, l'expression transformée est

$$(M, m)\ (N, n)\ (P, p)\ (Q, q)\ (R, r)\ \{1, 2, 4, 4, 2, 1\};$$

en représentant par cette notation abrégée la fonction

$$\begin{aligned} &1\,.\ MNPQR \\ &+ 2\Sigma\,(mNPQR) \\ &+ 4\Sigma\,(mn\,PQR) \\ &+ 4\Sigma\,(mn\,p\,QR) \\ &+ 2\Sigma\,(mn\,p\,qR) \\ &+ 1\,.\ mn\,p\,q\,r. \end{aligned}$$

En écartant les relations $M = m^2 - m, \ldots$, et en supposant seulement que $(m, n, p, q, r)$ soient les ordres, et $(M, N, P, Q, R)$ les classes des cinq courbes, la nouvelle formule s'applique aux courbes avec des points doubles ou de rebroussement; on peut même supposer que la courbe de la classe $M$ et de l'ordre $m$ se réduise à un système de $M$ points et de $m$ droites, et que les autres courbes se réduisent aussi à des systèmes de points et droites; et cela étant, on obtient une vérification immédiate de la formule. Car en choisissant dans le système qui remplace chaque courbe un élément (point ou droite) à volonté, on obtient

$$\begin{array}{rl} MNPQR & \text{systèmes } 5p, \\ \Sigma\, m\, NPQR & \text{systèmes } 4p,\ 1d. \\ \Sigma\, m\, n\, PQR & \text{systèmes } 3p,\ 2d, \\ \Sigma\, m\, n\, p\, QR & \text{systèmes } 2p,\ 3d, \\ \Sigma\, m\, n\, p\, q\, R & \text{systèmes } 1p,\ 4d, \\ m\, n\, p\, q\, r & \text{systèmes } \quad 5d. \end{array}$$

Or la condition par rapport à la première courbe se réduit à celle de passer par l'un quelconque des $M$ points, ou de toucher l'une quelconque des $m$ droites; et de même pour les autres courbes. Donc (en entendant par le mot *toucher* appliqué à un système de points et de droites, passer par les points et toucher les droites du système) la conique doit toucher l'un quelconque des systèmes ($5p$), ou ($4p$, $1d$), ou ($3p$, $2d$), ..., ou ($5d$); et pour un système de la forme

$$(5p),\quad (4p,\ 1d),\quad (3p,\ 2d),\quad (2p,\ 3d),\quad (1p,\ 4d),\quad \text{ou}\quad (5d),$$

le nombre des coniques est

$$1,\qquad 2,\qquad 4,\qquad 4,\qquad 2,\qquad \text{ou}\quad 1,$$

ce qui donne pour le nombre total des coniques l'expression ci-dessus écrite.

On peut supposer que la conique, au lieu de toucher les deux courbes $m$, $n$, ait avec la seule courbe $m$ (1°) un contact du deuxième ordre; (2°) un contact double. Le nombre des coniques qui satisfont à l'une ou l'autre de ces conditions, et qui passent aussi par trois points donnés, a été trouvé par Steiner (Aufgabe und Lehrsätze, *Crelle*, t. XLIX. p. 273), savoir:

$$(1°)\ \text{le nombre} = 3m(m-1);\quad (2°)\ \text{le nombre} = \tfrac{1}{2}(m^2-m)(m^2+3m-6);$$

j'ai vérifié d'une manière particulière ces deux résultats.

En supposant que la conique (au lieu de passer par les trois points donnés) touche les courbes $p$, $q$, $r$, je trouve pour les deux cas respectivement ces résultats,

1° le nombre $= 3(m^2-m) \times (P,\ p)(Q,\ q)(R,\ r)\{1,\ 2,\ 2,\ 1\}$,

2° le nombre $= \tfrac{1}{2}(m^2-m) \times$

$$(P,\ p)(Q,\ q)(R,\ r)\{m^2+3m-6,\ 2m^2+6m-16,\ 4m^2+4m-22,\ 4m^2-15\};$$

formules dans lesquelles la courbe $m$ doit être une courbe sans singularités.

*Grasmere, Westmoreland*, 37 *Juillet*, 1864.

# 307.

## NOTE SUR LES FONCTIONS al $(x)$, &c., DE M. WEIERSTRASS.

[From the *Journal de Mathématiques* (Liouville), tom. VII. (1862), pp. 137—142.]

Les fonctions al $(x)$, al $(x)_1$, al $(x)_2$, al $(x)_3$ de M. Weierstrass satisfont respectivement aux équations

$$\frac{d^2 \operatorname{al}(x)}{dx^2} + 2k^2x\frac{d \operatorname{al}(x)}{dx} + 2kk'^2\frac{d \operatorname{al}(x)}{dk} + \qquad k^2x^2 \operatorname{al}(x) = 0,$$

$$\frac{d^2 \operatorname{al}(x)_1}{dx^2} + 2k^2x\frac{d \operatorname{al}(x)_1}{dx} + 2kk'^2\frac{d \operatorname{al}(x)_1}{dk} + (k'^2 + k^2x^2) \operatorname{al}(x)_1 = 0,$$

$$\frac{d^2 \operatorname{al}(x)_2}{dx^2} + 2k^2x\frac{d \operatorname{al}(x)_2}{dx} + 2kk'^2\frac{d \operatorname{al}(x)_2}{dk} + (1 \ + k^2x^2) \operatorname{al}(x)_2 = 0,$$

$$\frac{d^2 \operatorname{al}(x)_3}{dx^2} + 2k^2x\frac{d \operatorname{al}(x)_3}{dx} + 2kk'^2\frac{d \operatorname{al}(x)_3}{dk} + (k^2 \ + k^2x^2) \operatorname{al}(x)_3 = 0,$$

ou, ce qui est la même chose, les fonctions

$$\operatorname{al}(x), \quad \sqrt{k} \operatorname{al}(x)_1, \quad \frac{\sqrt{k}}{\sqrt{k'}} \operatorname{al}(x)_2, \quad \frac{1}{\sqrt{k'}} \operatorname{al}(x)_3,$$

satisfont chacune à l'équation

$$\frac{d^2z}{dx^2} + 2k^2x\frac{dz}{dx} + 2kk'^2\frac{dz}{dk} + k^2x^2z = 0.$$

Écrivons pour un moment $\xi$, $\kappa$ au lieu de $x$, $k$; les fonctions al $(\xi)$, etc., satisfont à l'équation

$$\frac{d^2z}{d\xi^2} + 2\kappa^2\xi\frac{dz}{d\xi} + 2\kappa\kappa'^2\frac{dz}{d\kappa} + \kappa^2\xi^2z = 0.$$

Cela étant, en posant

$$\xi = \frac{x}{\sqrt{k}}, \quad \kappa = k,$$

on obtient

$$\frac{dz}{dx} = \frac{1}{\sqrt{k}}\frac{dz}{d\xi}, \quad \frac{d^2z}{dx^2} = \frac{1}{k}\frac{d^2z}{d\xi^2}, \quad \frac{dz}{dk} = -\frac{x}{2k\sqrt{k}}\frac{dz}{d\xi} + \frac{dz}{d\kappa},$$

et de là

$$\frac{dz}{d\xi} = \sqrt{k}\frac{dz}{dx}, \quad \frac{d^2z}{d\xi^2} = k\frac{d^2z}{dx^2}, \quad \frac{dz}{d\kappa} = \frac{dz}{dk} + \frac{x}{2k}\frac{dz}{dx}.$$

L'équation différentielle devient ainsi

$$k\frac{d^2z}{dx^2} + \quad 2k^2x\frac{dz}{dx} + 2kk'^2\left(\frac{dz}{dk} + \frac{x}{2k}\frac{dz}{dx}\right) + kx^2z = 0,$$

c'est-à-dire

$$\frac{d^2z}{dx^2} + \frac{1+k^2}{k}x\frac{dz}{dx} + \quad 2k'^2\frac{dz}{dk} \qquad + \quad x^2z = 0,$$

équation qui sera ainsi satisfaite par

$$\mathrm{al}\left(\frac{x}{\sqrt{k}}\right), \quad \sqrt{k}\,\mathrm{al}\left(\frac{x}{\sqrt{k}}\right)_1, \quad \frac{\sqrt{k}}{\sqrt{k'}}\mathrm{al}\left(\frac{x}{\sqrt{k}}\right)_2, \quad \frac{1}{\sqrt{k'}}\mathrm{al}\left(\frac{x}{\sqrt{k}}\right)_3.$$

Or en écrivant

$$k + \frac{1}{k} = \alpha,$$

l'équation en $z$ devient

$$(1) \qquad \frac{d^2z}{dx^2} + 2\alpha x\frac{dz}{dx} - 2(\alpha^2 - 4)\frac{dz}{d\alpha} + x^2z = 0,$$

laquelle est ce que devient celle-ci

$$(2) \quad (1 - \alpha x^2 + x^4)\frac{d^2z}{dx^2} + (n-1)(\alpha x - 2x^3)\frac{dz}{dx} - 2n(\alpha^2 - 4)\frac{dz}{d\alpha} + n(n-1)x^2z = 0,$$

en y écrivant $\frac{x}{\sqrt{n}}$ au lieu de $x$, et puis $n = \infty$. L'équation (2), trouvée par Jacobi (*Journal de Crelle*, t. IV. p. 185, 1829), a la propriété que voici, savoir en posant

$$\alpha = k + \frac{1}{k}, \quad x = \sqrt{k}\sin\mathrm{am}\, u,$$

alors l'équation est satisfaite en prenant pour $z$ soit le numérateur, soit le dénominateur, de la fonction rationnelle de $x$ qui donne la valeur de la fonction $\sqrt{\lambda}\sin\mathrm{am}\left(\frac{\mu}{M}, \lambda\right)$, où $\lambda$, $M$ sont le module et le multiplicateur qui correspondent à la transformation de l'ordre $n$ ($n$ étant un nombre impair quelconque).

L'équation fut donnée par Jacobi sans démonstration. Je l'ai démontrée (*Camb. and Dubl. Math. Journal*, t. II. p. 256, 1847, [45]) de la manière que voici, savoir en écrivant

$$z = \left(\frac{2Kk'}{\pi}\right)^{\frac{1}{2}(n-1)} \Theta^{-n}(u) \,.\, \Sigma,$$

on obtient pour $\Sigma$ l'équation

$$\frac{d^2\Sigma}{du^2} - 2nu\left(k'^2 - \frac{E}{K}\right)\frac{d\Sigma}{du} + 2nkk'^2\frac{d\Sigma}{dk} = 0.$$

Cette équation, en prenant

$$\omega = \frac{n\pi K'}{K}, \quad v = \frac{n\pi u}{2K},$$

devient

$$\frac{d^2\Sigma}{dv^2} - 4\frac{d\Sigma}{d\omega} = 0,$$

équation mentionnée par Jacobi, laquelle est satisfaite par

$$\Sigma = \Theta\left(nu, \frac{nK'}{K}\right), \text{ ou } \Sigma = \mathrm{H}\left(nu, \frac{nK'}{K}\right);$$

cela donne pour $z$ deux valeurs qui sont le dénominateur et le numérateur de la fraction dont il s'agit. J'ose croire que ce doit être à peu près de cette manière que l'équation fut trouvée par Jacobi.

Or les solutions en question de l'équation (1) peuvent être trouvées au moyen de l'équation de Jacobi; pour cela, au lieu des valeurs ci-dessus données de $\omega$, $v$, j'écris

$$\omega = \frac{\pi K'}{K}, \quad v = \frac{\sqrt{n}\pi u}{2K},$$

ce qui conduit à la même équation

$$\frac{d^2\Sigma}{dv^2} - 4\frac{d\Sigma}{d\omega} = 0,$$

laquelle sera ainsi satisfaite par

$$\Sigma = \Theta\left(\sqrt{n}u, \frac{K'}{K}\right), \quad \Sigma = \mathrm{H}\left(\sqrt{n}u, \frac{K'}{K}\right),$$

ou, ce qui est la même chose, par

$$\Sigma = \Theta(\sqrt{n}u), \qquad \Sigma = \mathrm{H}(\sqrt{n}u).$$

L'équation (2) sera donc satisfaite par

$$z = \left(\frac{2Kk'}{\pi}\right)^{\frac{1}{2}(n-1)} \Theta^{-n}(u)\, \Theta(\sqrt{n}u),$$

ou, en se souvenant que $\Theta(0)=\sqrt{\dfrac{2Kk'}{\pi}}$, par

$$z=\frac{\Theta(\sqrt{n}u)}{\Theta(0)}\cdot\frac{\Theta^n(0)}{\Theta^n(u)}.$$

Or en écrivant $\dfrac{x}{\sqrt{n}}$ au lieu de $x$, pour faire ensuite $n=\infty$, nous avons

$$\frac{x}{\sqrt{n}}=\sqrt{k}\sin\operatorname{am}u,$$

équation qui se réduit à

$$u=\frac{x}{\sqrt{nk}};$$

cela donne

$$\Theta(\sqrt{n}u)=\Theta\left(\frac{x}{\sqrt{k}}\right),\quad \Theta^n u=\Theta^n(0)\,e^{\frac{1}{2}nu^2\left(1-\frac{E}{K}\right)}=\Theta^n(0)\,e^{\frac{x^2}{2k}\left(1-\frac{E}{K}\right)},$$

puisque

$$\Theta(u)=\Theta(0)\,e^{\frac{1}{2}u^2\left(1-\frac{E}{K}\right)-k^2\int_0 du\int_0 du\sin^2\operatorname{am}u};$$

et $n\displaystyle\int_0 du\int_0 du\sin^2\operatorname{am}u$, en y substituant $u=\dfrac{x}{\sqrt{nk}}$, contient le facteur $\dfrac{1}{n}$ et se réduit ainsi à zéro. Donc on obtient

$$z=\frac{\Theta\left(\frac{x}{\sqrt{k}}\right)}{\Theta(0)}\,e^{\frac{x^2}{2k}\left(1-\frac{E}{K}\right)}$$

comme solution de l'équation (1), qui se déduit de l'équation (2) en y écrivant $\dfrac{x}{\sqrt{n}}$ au lieu de $x$ et puis $n=\infty$. Et cette valeur de $z$ est précisément la fonction al$\left(\dfrac{x}{\sqrt{k}}\right)$ de M. Weierstrass. On obtient de même la solution

$$z=\frac{\mathrm{H}\left(\frac{x}{\sqrt{k}}\right)}{\Theta(0)}\,e^{\frac{x^2}{2k}\left(1-\frac{E}{K}\right)},$$

ou, ce qui est la même chose,

$$z=\frac{\mathrm{H}\left(\frac{x}{\sqrt{k}}\right)}{\frac{1}{\sqrt{k}}\mathrm{H}'(0)}\,e^{\frac{x^2}{2k}\left(1-\frac{E}{K}\right)},$$

qui est la fonction $\sqrt{k}\,\mathrm{al}\left(\dfrac{x}{\sqrt{k}}\right)_1$. Et d'une manière semblable les solutions

$$z=\frac{\mathrm{H}\left(\frac{x}{\sqrt{k}}+K\right)}{\Theta(0)}\,e^{\frac{x^2}{2k}\left(1-\frac{E}{K}\right)},\quad z=\frac{\Theta\left(\frac{x}{\sqrt{k}}+K\right)}{\Theta(0)}\,e^{\frac{x^2}{2k}\left(1-\frac{E}{K}\right)}$$

qui sont les fonctions $\sqrt{\dfrac{k}{k'}}\,\mathrm{al}\left(\dfrac{x}{\sqrt{k}}\right)_2$, $\dfrac{1}{\sqrt{k'}}\,\mathrm{al}\left(\dfrac{x}{\sqrt{k}}\right)_3$.

J'ajoute que dans le Mémoire cité (1847) j'ai donné la suite

$$z = C_0 + C_1 \frac{x^2}{1.2} + C_2 \frac{x^4}{1.2.3.4} + \dots,$$

où

$$\begin{aligned}
C_0 &= 1,\\
C_1 &= 0,\\
C_2 &= -2,\\
C_3 &= +8\alpha,\\
C_4 &= -32\alpha^2 - 4,\\
C_5 &= +128\alpha^3 + 96\alpha,\\
C_6 &= -512\alpha^4 - 960\alpha^2 - 408,\\
C_7 &= +2048\alpha^5 + 7168\alpha^3 + 7584\alpha,\\
C_8 &= -8192\alpha^6 - 46080\alpha^4 - 88320\alpha^2 - 15384,\\
&\vdots
\end{aligned}$$

de façon qu'en général

$$C_{r+2} = -(2r+1)(2r+2)C_r - (2r+2)\alpha C_{r+1} + 2(\alpha^2-4)\frac{dC_{r+1}}{d\alpha},$$

c'est le développement de M. Weierstrass pour la fonction al $\left(\frac{x}{\sqrt{k}}\right)$: seulement je ne connaissais pas alors l'expression finie

$$\text{al}\left(\frac{x}{\sqrt{k}}\right), \quad \text{ou} \quad \frac{1}{\Theta(0)}\,\Theta\left(\frac{x}{\sqrt{k}}\right) e^{\frac{x^2}{2k}\left(1-\frac{E}{K}\right)}$$

de cette suite. Je remarque en passant que pour $\alpha = 2$, la suite se réduit à

$$e^{\frac{1}{2}x^2} . \tfrac{1}{2}(e^x + e^{-x}).$$

# 308.

## ON THE $\triangle$ FACED POLYACRONS, IN REFERENCE TO THE PROBLEM OF THE ENUMERATION OF POLYHEDRA.

[From the *Memoirs of the Literary and Philosophical Society of Manchester*, vol. I. (1862), pp. 248—256.]

THE problem of the enumeration of polyhedra [1] is one of extreme difficulty, and I am not aware that it has been discussed elsewhere than in Mr Kirkman's valuable series of papers on this subject in the *Memoirs* of the Society and in the *Philosophical Transactions*. A case of the general problem is that of the enumeration of the polyhedra with trihedral summits; and Mr Kirkman in the earliest of his papers, viz. that "On the representation and enumeration of polyhedra" (*Memoirs*, vol. XII. pp. 47—70, 1854), has in fact, by an examination of the particular case, accomplished the enumeration of the octahedra with trihedral summits. A subsequent paper "On the enumeration of $x$-edra having trihedral summits and an $(x-1)$gonal base," *Phil. Trans.* vol. XLVI. pp. 399—411, 1856), relates, as the title shows, only to a special case of the problem of the polyhedra with trihedral summits, and in this particular case the number of polyhedra is more completely determined; but the later memoirs relate to the problem in all its generality, and the above-mentioned particular problem of the enumeration of the polyhedra with trihedral summits is not, I think, any where resumed. Instead of the polyhedra with trihedral summits, it is really the same thing, but it is rather more convenient to consider the polyacrons with triangular faces, or as these may for shortness be called, the $\triangle$ faced polyacrons; and it is intended in the present paper to give a method for the derivation of the $\triangle$ faced polyacrons of a given number of summits from those of the next inferior number of summits, and to exemplify it by finding, in an orderly manner, the $\triangle$ faced polyacrons

[1] I use with Mr Kirkman the expression "enumeration of polyhedra" to designate the general problem, but I consider that the problem is to find the different polyhedra rather than to count them, and I consequently take the word *enumeration* in the popular rather than the mathematical sense.

up to the octacrons: thus, as regards the examples, stopping at the same point as Mr Kirkman, for although perfectly practicable it would be very tedious to carry them further, and there would be no commensurate advantage in doing so. The epithet $\Delta$ faced will be omitted in the sequel, but it is to be understood throughout that I am speaking of such polyacrons only; and I shall for convenience use the epithets tripleural, tetrapleural, &c. to denote summits with three, four, &c. edges through them. The number of edges at a summit is of course equal to the number of faces, but it is the edges rather than the faces which have to be considered.

An $n$-acron has

$$n \text{ summits, } 3n-6 \text{ edges, } 2n-4 \text{ faces,}$$

and it is easy to see that there are the following three cases only, viz.:

1. The polyacron has at least one tripleural summit.

2. The polyacron, having no tripleural summit, has at least one tetrapleural summit.

3. The polyacron, having no tripleural or tetrapleural summit, has at least twelve pentipleural summits.

In fact, if the polyacron has $c$ tripleural summits, $d$ tetrapleural summits, $e$ pentipleural summits, and so on, then we have

$$n = c + d + e + f + g + h + \&\text{c.},$$

$$6n - 12 = 3c + 4d + 5e + 6f + 7g + 8h + \&\text{c.},$$

and therefore

$$12 = 3c + 2d + e + 0f - g - 2h - \&\text{c.},$$

or

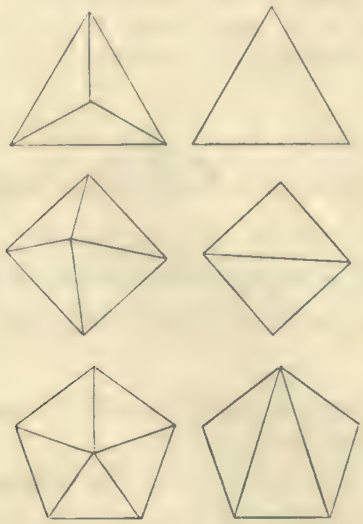
$$3c + 2d + e = 12 + g + 2h + \&\text{c.};$$

whence if $c=0$ and $d=0$, then $e=12$ at least. It appears, moreover (since $n$ cannot be less than $e$), that any polyacron with less than 12 summits cannot belong to the third class, and must therefore belong to the first or the second class.

An $(n+1)$-acron, by a process which I call the subtraction of a summit, may be reduced to an $n$-acron; viz., the faces about any summit of the $(n+1)$-acron stand upon a polygon (not in general a plane figure) which may be called the basic polygon, and when the summit with the faces and edges belonging to it is removed, the basic polygon, if a triangle, will be a face of the $n$-acron; if not a triangle, it can be partitioned into triangles which will be faces of the $n$-acron. The annexed figures exhibit the process for the cases of a tripleural, tetrapleural and pentipleural summit respectively, which are the only cases which need be considered; these may be called the first, second and third process respectively. It is proper to remark that for the same removed summit the first process can be performed in one way only, the second process in two ways, the third in five ways; these being in fact the numbers of ways of partitioning the basic polygon.

We may in like manner, by the converse process of the addition of a summit, convert an $n$-acron into an $(n+1)$-acron; viz., it is only necessary to take on the

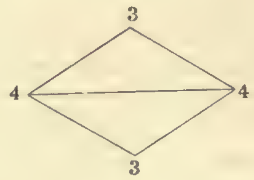
$n$-acron a polygon of any number of sides, and make this the basic polygon of the new summit of the $(n+1)$-acron, and for this purpose to remove the faces within the polygon and substitute for them a set of triangular faces standing on the sides of the polygon and meeting in the new summit: the same figures exhibit the process for the cases of a tripleural, tetrapleural and pentipleural summit respectively, which



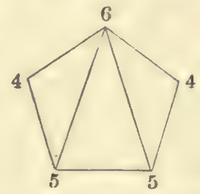
(as for the subtractions) are the only cases which need be considered. It may be noticed that for the same basic polygon the process is in each case a unique one; the process is said to be the first, second, or third process, according as the new summit is tripleural, tetrapleural, or pentipleural.

Now, reverting to the before-mentioned division of the polyacrons into three classes, an $(n+1)$-acron of the first class may by the first process of subtraction be reduced to an $n$-acron, and conversely it can be by the first process of addition derived from an $n$-acron. An $(n+1)$-acron of the second class, as having a tetrapleural summit, may by the second process of subtraction be reduced to an $n$-acron, and conversely it can be by the second process of addition derived from an $n$-acron. And in like manner, an $(n+1)$-acron of the third class, as having a pentipleural summit, may be by the third process of subtraction reduced to an $n$-acron, and conversely it may be by the third process of addition derived from an $n$-acron.

Hence all the $(n+1)$-acrons can be by the first, second and third processes of addition respectively derived from the $n$-acrons. It is to be observed that all the $(n+1)$-acrons of the first class are obtained by the first process; the second process is only required for finding the $(n+1)$-acrons of the second class; and these being all obtained by means of it, the third process is only required for finding the $(n+1)$-acrons of the third class. Hence the second process need only be made use of when the $n$-acron has no tripleural summit, or when it has only one tripleural summit, or when, having two tripleural summits, they are the opposite summits of two adjacent faces. In the last-mentioned two cases respectively it is only necessary to consider the basic quadrangles which pass through the single tripleural summit and the basic quadrangle which passes through the two tripleural summits; for with any other basic quadrangle the derived $(n+1)$-acron would retain a tripleural summit, and would consequently be of the first class. The condition is more simply expressed as follows, viz.: The second process need only be employed when there is on the $n$-acron a basic quadrangle the summits of which are at least of the number of edges shown in the




annexed figure, and all the other summits are at least 4-pleural. Again, by the third process (as already mentioned) we seek only to obtain the $(n+1)$-acrons of the third class; the process need only be applied to the $n$-acrons for which there exists a basic pentagon the summits of which are at least of the number of edges shown in the




annexed figure, all the other summits being at least 5-pleural; for it is only in this case that the derived $(n+1)$-acron will be of the third class. The condition just referred to obviously implies that the $n$-acron is of the second or third class. It is to be noticed that in applying the foregoing principles to the formation of the polyacrons as far as the 11-acrons we are only concerned with the first and second processes.

Consider the entire series of $n$-acrons, say $A$, $B$, $C$, &c., and suppose that the $n$-acron $A$ gives rise to a certain number, say $P$, $Q$, $R$, $S$ of $(n+1)$-acrons, the $(n+1)$-acron $P$ is of course derivable from the $n$-acron $A$, but it may be derivable from other $n$-acrons, suppose from the $n$-acrons $B$ and $C$. Then in considering the $(n+1)$-acrons derived from $B$, one of these will of course be found to be the $(n+1)$-acron $P$, and it is only the remaining $(n+1)$-acrons derived from $B$ which are or may be $(n+1)$-acrons not already previously obtained as $(n+1)$-acrons derived from $A$. And if in this manner, as soon as each $(n+1)$-acron is obtained, we apply to it the process of subtraction so as to ascertain the entire series of $n$-acrons from which it is derivable, and, in forming the $(n+1)$-acrons derived from these, take account of the $(n+1)$-acrons already previously obtained and found to be derivable from these, we should obtain without any repetitions the entire series of the $(n+1)$-acrons.

For merely finding the number of the $(n+1)$-acrons, a more simple process might be adopted: say that an $n$-acron is $p$-wise generating when it gives rise to a number $p$ of $(n+1)$-acrons, and that it is $q$-wise generable when it can be derived from a number $q$ of $(n+1)$-acrons; and assume that a given $n$-acron is $(y_1+y_2+y_3+\&c.)$-wise generating, viz. that it gives rise to a number $y_1$ of $(n+1)$-acrons which are 1-wise generable, a number $y_2$ of $(n+1)$-acrons which are 2-wise generable, and so on; these forming the sum

$$\Sigma\,(y_1+\tfrac{1}{2}y_2+\tfrac{1}{3}y_3+\ldots)$$

where $\Sigma$ refers to the entire series of the $n$-acrons, it is clear that every $m$-wise generable $(n+1)$-acron will in respect of each of the $n$-acrons from which it is derivable be reckoned as $\frac{1}{m}$, that is, it will be in the entire sum reckoned as 1, and the sum in question will consequently be the number of the $(n+1)$-acrons.

The figures of the polyacrons comprised in the annexed Tables show the application of the method to the genesis of the polyacrons as far as the octacrons, in which the numbers indicate the nature of the different summits, according to the number of edges through each summit, viz., 3 a tripleural summit, 4 a tetrapleural summit, and so on. It will be noticed that there is only a single case in which this notation is insufficient to distinguish the polyacron, viz., among the octacrons there are two forms each of them with the same symbol 33445566; the inspection of the figures shows at once that these are wholly distinct forms, for in the first of them, viz. that derived from 3344555, each of the tripleural summits stands upon a basic triangle 456, while in the other of them, that from 3444555, each of the tripleural summits stands upon a basic triangle 566. But the symbol is merely generic, and of course in the polyacrons of a greater number of summits it may very well happen that a considerable number of polyacrons are comprised in the same genus.

The following remarks on the derivation of the octacrons from the heptacrons will further illustrate the method:

1. The heptacron 3335556 has three kinds of faces, viz. 355[1], 356, 555, the first process consequently gives rise to 3 octacrons. As the heptacron has more than two tripleural summits the second process is not applicable.

2. The heptacron 3344466 has three kinds of faces, viz.: 366, 346 and 446, and the first process gives therefore 3 octacrons. The heptacron has only two tripleural summits, and they are disposed in the proper manner; the second process gives therefore 1 octacron.

3. The heptacron 3344556 has five kinds of faces, viz. 345, 346, 356, 456 and 455, and the first process consequently gives 5 octacrons. The heptacron has two tripleural summits, but they are not disposed in such manner as to render the second process applicable.

[1] It is hardly necessary to remark that it must not be imagined that in general all the faces denoted by a symbol such as 355 (which determines only the nature of the summits on the face) are faces of the same kind, but this is so in the cases referred to in the text.

4. The heptacron 3444555 has four kinds of faces, viz. 355, 455, 445 and 444, and the first process gives therefore 4 octacrons. The heptacron has one tripleural summit, and the basic quadrangles 3545 which belong to it are of the same kind; the second process gives therefore 1 octacron.

5. The heptacron 4444455 has only one kind of face, viz. 445, and the first process gives therefore 1 octacron. There are two kinds of basic quadrangles, viz. 4545 and 4445, and the second process gives therefore 2 octacrons.

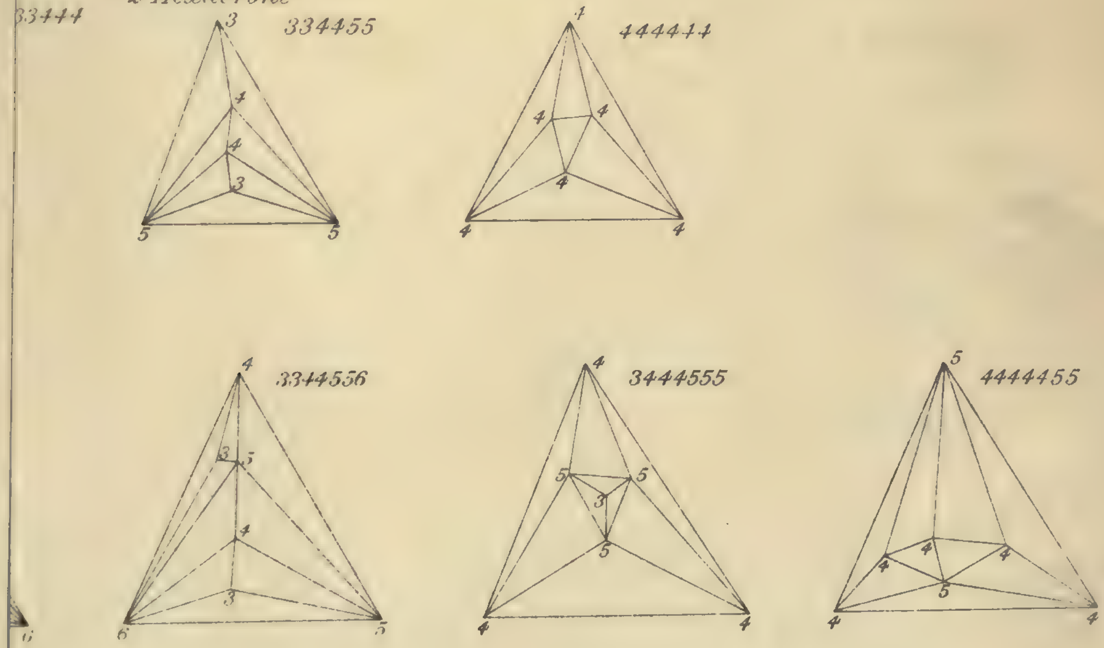
The number of octacrons would thus be 20, but by passing back from the octacrons to the heptacrons, it is found that there are in fact only 14 octacrons. Thus the octacron 33336666 has only one kind of tripleural summit 666 (the summit is here indicated by the symbol of the basic polygon) and the octacron is thus seen to be derivable from a single heptacron only, viz. the heptacron 3335556 from which it was in fact derived. But the octacron 33345567 has three kinds of tripleural summits, viz. 567, 557 and 467, and it is consequently derivable from three heptacrons, viz. the heptacrons 3335556, 3344466 and 3344555, and so on. The passage to the heptacrons from an octacron with one or more tripleural summits is of course always by the first process, but for the last two octacrons, which have no tripleural summits, the passage back to the heptacrons is by the second process: thus for the octacron 44445555 we have but one kind of tetrapleural summit 4555; but as opposite pairs of summits of the basic quadrangle are of different kinds, viz. 45 and 55, we obtain two heptacrons, viz. 3444555 and 4444455. The octacron 44444466 has but one kind of tetrapleural summit, viz. 4646, and the pairs of opposite summits of the basic quadrangle being of the same kind 46, we obtain from it only the heptacron 4444455.

It may be remarked that for the five heptacrons respectively the values of the sum $y_1 + \frac{1}{2}y_2 + \frac{1}{3}y_3 + \dots$ are

$$1 + \tfrac{1}{3} + \tfrac{1}{2},\quad \tfrac{1}{3} + 1 + \tfrac{1}{2} + \tfrac{1}{2},\quad \tfrac{1}{3} + \tfrac{1}{2} + \tfrac{1}{2} + 1 + 1,\quad 1 + 1 + 1 + 1 + \tfrac{1}{2},\quad 1 + \tfrac{1}{2} + \tfrac{1}{2},$$

giving for $\Sigma\,(y_1 + \frac{1}{2}y_2 + \frac{1}{3}y_3 + \dots)$ the value 14, as it should do.


2 Hexacrons




To face p. 4

# 309.

## NOTE ON THE THEORY OF DETERMINANTS.

[From the *Philosophical Magazine*, vol. XXI. (1861), pp. 180—185.]

THE following mode of arrangement of the developed expression of a determinant had presented itself to me as a convenient one for the calculation of a rather complicated determinant of the fifth order; but I have since found that it is in effect given, although in a much less compendious form, in a paper by J. N. Stockwell, "On the Resolution of a System of Symmetrical Equations with Indeterminate Coefficients," Gould's *Ast. Journal*, No. 139 (Cambridge, U. S., Sept. 10, 1860).

Suppose that the determinant

$$\begin{vmatrix} 11, & 12, & 13 \\ 21, & 22, & 23 \\ 31, & 32, & 33 \end{vmatrix}$$

is represented by $\{123\}$, and so for a determinant of any order $\{123 \ldots n\}$.

Let $|1|$, $|2|$, $|12|$, $|123|$, &c., denote as follows: viz.

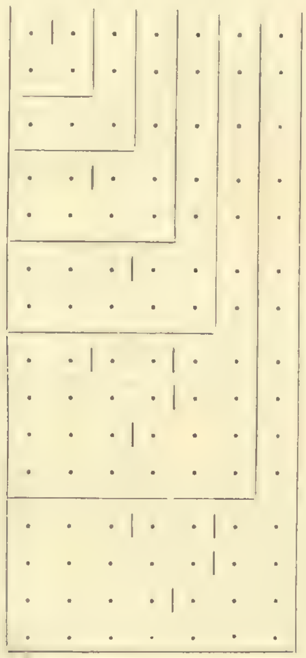
$$\begin{aligned} |1| &= 11, \quad |2| = 22, \text{ \&c.} \\ |12| &= 12 . 21, \\ |123| &= 12 . 23 . 31, \\ &\text{\&c.,} \end{aligned}$$

where it is to be noticed that, with the same two symbols, e.g. 1 and 2, there is but one distinct expression $|12|$ (in fact $|21| = 21 . 12 = |12|$); with the same three symbols, 1, 2, 3, there are two distinct expressions, $|123|$ $(=12 . 23 . 31)$ and $|132|$ $(=13 . 32 . 21)$; and generally with the same $m$ symbols $1, 2, 3 \ldots m$, there are $1 . 2 . 3 \ldots (m-1)$ distinct expressions $\vert\, 123 \ldots m \,\vert$, which are obtained by permuting in every possible manner all but one of the $m$ symbols.

This being so, and writing for greater simplicity $\vert\, 1 \,\vert\, 2 \,\vert$ to denote the product $\vert\, 1 \,\vert \times \vert\, 2 \,\vert$, and so in general, the values of the determinants $\{12\}$, $\{123\}$, $\{1234\}$, $\{12345\}$, &c. are as follows: viz.

$$\begin{array}{rl|cc}
 & & \multicolumn{2}{c}{\text{No. of terms.}} \\
 & & + & - \\
\{12\} = & + \vert 1 \vert 2 \vert \;\ldots\ldots & 1 & \\
 & - \vert 1 \;\; 2 \vert \;\ldots\ldots & & 1 \\
\hline
 & & \multicolumn{2}{c}{1 + 1 = 2} \\
\hline
\{123\} = & + \vert 1 \vert 2 \vert 3 \vert \;\ldots & 1 & \\
 & - \vert 1 \;\; 2 \vert 3 \vert \;\ldots & & 3 \\
 & + \vert 1 \;\; 2 \;\; 3 \vert \;\ldots & 2 & \\
\hline
 & & \multicolumn{2}{c}{3 + 3 = 6} \\
\hline
\{1234\} = & + \vert 1 \vert 2 \vert 3 \vert 4 \vert \;\ldots & 1 & \\
 & - \vert 1 \;\; 2 \vert 3 \vert 4 \vert \;\ldots & & 6 \\
 & + \vert 1 \;\; 2 \;\; 3 \vert 4 \vert \;\ldots & 8 & \\
 & + \vert 1 \;\; 2 \vert 3 \;\; 4 \vert \;\ldots & 3 & \\
 & - \vert 1 \;\; 2 \;\; 3 \;\; 4 \vert \;\ldots & & 6 \\
\hline
 & & \multicolumn{2}{c}{12 + 12 = 24} \\
\hline
\{12345\} = & + \vert 1 \vert 2 \vert 3 \vert 4 \vert 5 \vert \;. & 1 & \\
 & - \vert 1 \;\; 2 \vert 3 \vert 4 \vert 5 \vert \;. & & 10 \\
 & + \vert 1 \;\; 2 \;\; 3 \vert 4 \vert 5 \vert \;. & 20 & \\
 & + \vert 1 \;\; 2 \vert 3 \;\; 4 \vert 5 \vert \;. & 15 & \\
 & - \vert 1 \;\; 2 \;\; 3 \;\; 4 \vert 5 \vert \;. & & 30 \\
 & - \vert 1 \;\; 2 \;\; 3 \vert 4 \;\; 5 \vert \;. & & 20 \\
 & + \vert 1 \;\; 2 \;\; 3 \;\; 4 \;\; 5 \vert \;. & 24 & \\
\hline
 & & \multicolumn{2}{c}{60 + 60 = 120}
\end{array}$$

where, as regards the signs, it is to be observed that there is a sign $-$ for each compartment $\vert \quad \vert$ containing an even number of symbols; thus in the expression for $\{1234\}$, the terms $\vert\, 1\ 2 \,\vert\, 3\ 4 \,\vert$ have the sign $- - = +$, and the terms $\vert\, 1\ 2\ 3\ 4 \,\vert$ the sign $-$. Or, what comes to the same thing; when $n$ is even, the sign is $+$ or $-$ according as the number of compartments is even or odd; and contrariwise when $n$ is odd. As regards the remaining part of the expression, this merely exhibits the partitions of a set of $n$ things; and the formulæ for the several determinants up to the determinant of a given order are all of them obtained by means of the form



which is carried up to the order 7, but which can be further extended without any difficulty whatever.

It is perhaps hardly necessary, but I give at full length the expressions of the determinant of the third order: this is

$$\{123\} = \begin{array}{l} \quad\; |\,1\,|\,2\,|\,3\,| \\ \left.\begin{array}{l} -\,|\,1\;\;2\,|\,3\,| \\ -\,|\,2\;\;3\,|\,1\,| \\ -\,|\,3\;\;1\,|\,2\,| \end{array}\right\} \\ \left.\begin{array}{l} +\,|\,1\;\;2\;\;3\,| \\ +\,|\,1\;\;3\;\;2\,| \end{array}\right\}; \end{array}$$

and by writing down in like manner the expression for the twenty-four terms of the determinant of the fourth order, the notation will become perfectly clear.

The formula hardly requires a demonstration. The terms of a determinant $\{123\ldots n\}$, for example the determinant $\{1234\}$, are obtained by permuting in every possible manner the symbols in either column, say the second column, of the arrangement

$$\begin{array}{cc} 1 & 1 \\ 2 & 2 \\ 3 & 3 \\ 4 & 4 \end{array}$$

and prefixing the sign (+ or −) of the arrangement; and the resulting arrangements, for instance

$$\begin{array}{ccc} +\,1\ 1, & -\,1\ 2, & -\,1\ 2, \\ 2\ 2 & 2\ 1 & 2\ 3 \\ 3\ 3 & 3\ 3 & 3\ 4 \\ 4\ 4 & 4\ 4 & 4\ 1 \end{array}$$

are interpreted either into $+11.22.33.44$, $-12.21.33.44$, $-12.23.34.41$, or in the notation of the formula, into

$$+\,|\,1\,|\,2\,|\,3\,|\,4\,|,\quad -\,|\,12\,|\,3\,|\,4\,|,\quad -\,|\,1\,2\,3\,4\,|;$$

and so in general.

Suppose that any partition of $n$ contains $\alpha$ compartments each of $a$ symbols, $\beta$ compartments each of $b$ symbols ... ($a$, $b$, ... being all of them different and greater than unity), and $\rho$ compartments each of a single symbol, we have

$$n = \alpha a + \beta b + \ldots + \rho;$$

and writing, as usual, $\Pi a = 1.2.3\ldots a$, &c., the number of ways in which the symbols 1, 2, 3, ... $n$, can be so arranged in compartments is

$$\frac{\Pi n}{(\Pi a)^\alpha\,(\Pi b)^\beta\ldots\Pi\alpha\,\Pi\beta\ldots\Pi\rho};$$

but each such arrangement gives $(\Pi\,(a-1))^\alpha.(\Pi\,(b-1))^\beta$ terms of the determinant, and the corresponding number of terms therefore is

$$\frac{\Pi n}{a^\alpha b^\beta\ldots\Pi\alpha\,\Pi\beta\ldots\Pi\rho}.$$

The whole number of terms of the determinant is $\Pi n$, and we have thus the theorem

$$1 = \Sigma\,\frac{1}{a^\alpha b^\beta\ldots\Pi\alpha\,\Pi\beta\ldots\Pi\rho},$$

in which the summation corresponds to all the different partitions $n = \alpha a + \beta b, \ldots + \rho$, where $a$, $b$, ... are all of them different and greater than unity; a theorem given in Cauchy's *Mémoire sur les Arrangements* &c., 1844. But it is to be noticed also that, the number of the positive and negative terms being equal, we have besides

$$0 = \Sigma\,\frac{(-)^{\alpha(a-1)+\beta(b-1)+\ldots}}{a^\alpha b^\beta\ldots\Pi\alpha\,\Pi\beta\ldots\Pi\rho};$$

or, what is the same thing,

$$0 = \Sigma\,\frac{(-)^{n-\alpha-\beta\ldots-\rho}}{a^\alpha b^\beta\ldots\Pi\alpha\,\Pi\beta\,.\,.\,\Pi\rho};$$

and thence also

$$\tfrac{1}{2} = \Sigma\,\frac{1}{a^\alpha b^\beta\ldots\Pi\alpha\,\Pi\beta\ldots\Pi\rho};$$

where, as before, $n = \alpha a + \beta b \ldots + \rho \,(a, b, \ldots$ being all different and greater than unity); but the summation is restricted either to the partitions for which $n - \alpha - \beta \ldots - \rho$ is even, or else to those for which $n - \alpha - \beta \ldots - \rho$ is odd.

The formula affords a proof of the fundamental property of skew symmetrical determinants. In such a determinant we have not only $12 = -21$, &c., but also $11 = 0$, &c. Suppose that $n$, the order of the determinant, is odd; then in each line of the expression

$$\{123 \ldots n\} = | \, 1 \, | \, 2 \, | \ldots | \, n \, | \\ \pm \text{ \&c.}$$

of the determinant, there is at least one compartment $| \, 1 \, |$ or $| \, 123 \, |$ &c. containing an odd number of symbols: let $| \, 123 \, |$ be such a compartment, then the determinant contains the terms $| \, 123 \, | \, P$ and $| \, 132 \, | \, P$ (where $P$ represents the remaining compartments), that is, $12 \,.\, 23 \,.\, 31 \,.\, P$ and $13 \,.\, 32 \,.\, 21 \,.\, P$. But in virtue of the relations $12 = -21$, &c., we have

$$12 \,.\, 23 \,.\, 31 = -13 \,.\, 32 \,.\, 21;$$

and so in all similar cases; that is, the terms destroy each other, or the skew symmetrical determinant of an odd order is equal to zero.

The like considerations show that a skew symmetrical determinant of an even order is a perfect square. In fact, considering for greater simplicity the case $n = 4$, any line in the foregoing expression of $\{1234\}$ for which a compartment contains an odd number of symbols, gives rise to terms which destroy each other, and may be omitted. The expression thus reduces itself to

$$\begin{aligned} \{1234\} &= + \, | \, 12 \, | \, 34 \, | \quad 3 \text{ terms} \\ &\quad - | \, 12 \quad 34 \, | \quad 6 \text{ terms,} \end{aligned}$$

which is in fact the square of

$$12 \,.\, 34 + 13 \,.\, 42 + 14 \,.\, 23;$$

for the square of a term, say $12 \,.\, 34$, is $\overline{12}^2 \,.\, \overline{34}^2$ or $12 \,.\, 21 \,.\, 34 \,.\, 43$, that is, $| \, 12 \, | \, 34 \, |$, and the double of the product of two terms, say $12 \,.\, 34$ and $13 \,.\, 42$, is $2 \,.\, 12 \,.\, 34 \,.\, 13 \,.\, 42$, or $-12 \,.\, 24 \,.\, 43 \,.\, 31 - 13 \,.\, 34 \,.\, 42 \,.\, 21$, that is $- | \, 1243 \, | - | \, 1342 \, |$, and so for the other similar terms, and we have

$$\{1234\} = (12 \,.\, 34 + 13 \,.\, 42 + 14 \,.\, 23)^2:$$

and so in general, $n$ being any even number, the skew symmetrical determinant $\{123 \ldots n\}$ is equal to the square of the Pfaffian $12 \ldots n$, where the law of these Pfaffian functions is

$$\begin{aligned} 1234 \;\; &= 12 \,.\, 34 \;\; + 13 \,.\, 42 \;\; + 14 \,.\, 23 \\ 123456 &= 12 \,.\, 3456 + 13 \,.\, 4562 + 14 \,.\, 5623 + 15 \,.\, 6234 + 16 \,.\, 2345, \end{aligned}$$

where, in the second equation, 3456, &c. are Pfaffians, viz. $3456 = 34 \,.\, 56 + 35 \,.\, 64 + 36 \,.\, 45$; and so on.

2, *Stone Buildings, W.C., December* 28, 1860.

# 310.

## NOTE ON MR JERRARD'S RESEARCHES ON THE EQUATION OF THE FIFTH ORDER.

[From the *Philosophical Magazine*, vol. XXI. (1861), pp. 210—214.]

FUNCTIONS of the same set of quantities which are, by any substitution whatever, simultaneously altered or simultaneously unaltered, may be called *homotypical*. Thus all symmetric functions of the same set of quantities are homotypical: $(x+y-z-w)^2$ and $xy+zw$ are homotypical, &c.

It is one of the most beautiful of Lagrange's discoveries in the theory of equations, that, given the value of any function of the roots, the value of any homotypical function may be rationally determined[1]: in other words, that any homotypical function whatever is a rational function of the coefficients of the equation and of the given function of the roots.

The researches of Mr Jerrard are contained in his work, *An Essay on the Resolution of Equations*, London, Taylor and Francis, 1859. The solution of an equation of the fifth order is made to depend on an equation of the sixth order in $W$; and he conceives that he has shown that one of the roots of this equation is a rational function of another root: "The equation for $W$ will therefore belong to a class of equations of the sixth degree, the resolution of which can, as Abel has shown, be effected by means of equations of the second and third degrees; whence I infer the possibility of solving any proposed equation of the fifth degree by a finite combination of radicals and rational functions."

[1] The *à priori* demonstration shows the cases of failure. Suppose that the roots of a biquadratic equation are 1, 3, 5, 9; then, given $a+b=8$, we know that either $a=3$, $b=5$, or else $a=5$, $b=3$, and in either case $ab=15$; hence in the present case (which represents the general case), $a+b$ being known, the homotypical function $ab$ is rationally determined. But if the roots are 1, 3, 5, 7 (where $1+7=3+5$), then, given $a+b=8$, this is satisfied by ($a$, $b=3$, 5) or by ($a$, $b=1$, 7), and the conclusion is $ab=15$ or 7; so that here $ab$ is determined, not as before, rationally, but by a quadratic equation.

The above property of rational expressibility, if true for $W$, will be true for any function homotypical with $W$; and conversely. I proceed to inquire into the form of the function $W$.

The function $W$ is derived from the function $P$, which denotes any one of the quantities $p_1$, $p_2$, $p_3$. And if $x_1$, $x_2$, $x_3$, $x_4$, $x_5$ are the roots of the given equation of the fifth order, and if $\alpha$, $\beta$, $\gamma$, $\delta$, $\epsilon$ represent in an undetermined or arbitrary order of succession the five indices 1, 2, 3, 4, 5, and if $\iota$ denote an imaginary fifth root of unity (I conform myself to Mr Jerrard's notation), then $p_1$, $p_2$, $p_3$, and the other auxiliary quantities $t$, $u$, are obtained from the system of equations:

$$\begin{aligned}
x_\alpha^3 + p_1 x_\alpha^2 + p_2 x_\alpha + p_3 &= t + u,\\
x_\beta^3 + p_1 x_\beta^2 + p_2 x_\beta + p_3 &= \iota t + \iota^4 u,\\
x_\gamma^3 + p_1 x_\gamma^2 + p_2 x_\gamma + p_3 &= \iota^2 t + \iota^3 u,\\
x_\delta^3 + p_1 x_\delta^2 + p_2 x_\delta + p_3 &= \iota^3 t + \iota^2 u,\\
x_\epsilon^3 + p_1 x_\epsilon^2 + p_2 x_\epsilon + p_3 &= \iota^4 t + \iota u.
\end{aligned}$$

If from these equations we seek for the values of $p_1$, $p_2$, $p_3$, $t$, $u$, we have

$$1 : p_1 : p_2 : p_3 : -t : -u = \Pi_1 : \Pi_2 : \Pi_3 : \Pi_4 : \Pi_5 : \Pi_6,$$

where $\Pi_1$, $\Pi_2$, .. denote the determinants formed out of the matrix

$$\left|\begin{matrix}
x_\alpha^3, & x_\alpha^2, & x_\alpha, & 1, & 1, & 1\\
x_\beta^3, & x_\beta^2, & x_\beta, & 1, & \iota, & \iota^4\\
x_\gamma^3, & x_\gamma^2, & x_\gamma, & 1, & \iota^2, & \iota^3\\
x_\delta^3, & x_\delta^2, & x_\delta, & 1, & \iota^3, & \iota^2\\
x_\epsilon^3, & x_\epsilon^2, & x_\epsilon, & 1, & \iota^4, & \iota
\end{matrix}\right|,$$

i.e., denoting the columns of this matrix by 1, 2, 3, 4, 5, 6, we have $\Pi_1 = 23456$, $\Pi_2 = -34561$, $\Pi_3 = 45612$, &c. In particular, the value of $\Pi_1$ is

$$= \left|\begin{matrix}
x_\alpha^2, & x_\alpha, & 1, & 1, & 1\\
x_\beta^2, & x_\beta, & 1, & \iota, & \iota^4\\
x_\gamma^2, & x_\gamma, & 1, & \iota^2, & \iota^3\\
x_\delta^2, & x_\delta, & 1, & \iota^3, & \iota^2\\
x_\epsilon^2, & x_\epsilon, & 1, & \iota^4, & \iota
\end{matrix}\right|,$$

and developing, and putting for shortness $\{\alpha\beta\} = x_\alpha x_\beta (x_\alpha - x_\beta)$, &c., we have

$$\begin{aligned}
\Pi_1 = (\{\alpha\beta\} + \{\beta\gamma\} + \{\gamma\delta\} + \{\delta\epsilon\} + \{\epsilon\alpha\})(-2\iota + \iota^2 - \iota^3 + 2\iota^4)\\
+ (\{\alpha\gamma\} + \{\gamma\epsilon\} + \{\epsilon\beta\} + \{\beta\delta\} + \{\delta\alpha\})(+\iota + 2\iota^2 - 2\iota^3 - 2\iota^4);
\end{aligned}$$

and this is also the form of the other determinants, the only difference being as to the meaning of the symbol $\{\alpha\beta\}$, which, however, in each case denotes a function such that $\{\alpha\beta\} = -\{\beta\alpha\}$. Writing for greater shortness,

$$\{\alpha\beta\gamma\delta\epsilon\} = \{\alpha\beta\} + \{\beta\gamma\} + \{\gamma\delta\} + \{\delta\epsilon\} + \{\epsilon\alpha\},$$

and in like manner

$$\{\alpha\gamma\epsilon\beta\delta\} = \{\alpha\gamma\} + \{\gamma\epsilon\} + \{\epsilon\beta\} + \{\beta\delta\} + \{\delta\alpha\},$$

$\Pi_1$ is an unsymmetric linear function (without constant term) of $\{\alpha\beta\gamma\delta\epsilon\}$, $\{\alpha\gamma\epsilon\beta\delta\}$; or, what is all that is material, it is an unsymmetric function, containing only odd powers, of $\{\alpha\beta\gamma\delta\epsilon\}$, $\{\alpha\gamma\epsilon\beta\delta\}$.

If for

$$\alpha \quad \beta \quad \gamma \quad \delta \quad \epsilon$$

we substitute any one of the five arrangements

$$\begin{array}{ccccc} \alpha & \beta & \gamma & \delta & \epsilon, \\ \beta & \gamma & \delta & \epsilon & \alpha, \\ \gamma & \delta & \epsilon & \alpha & \beta, \\ \delta & \epsilon & \alpha & \beta & \gamma, \\ \epsilon & \alpha & \beta & \gamma & \delta, \end{array}$$

then $\{\alpha\beta\gamma\delta\epsilon\}$ and $\{\alpha\gamma\epsilon\beta\delta\}$ will in each case remain unaltered.

But if we substitute any one of the five arrangements

$$\begin{array}{ccccc} \alpha & \epsilon & \delta & \gamma & \beta, \\ \epsilon & \delta & \gamma & \beta & \alpha, \\ \delta & \gamma & \beta & \alpha & \epsilon, \\ \gamma & \beta & \alpha & \epsilon & \delta, \\ \beta & \alpha & \epsilon & \delta & \gamma, \end{array}$$

then in each case $\{\alpha\beta\gamma\delta\epsilon\}$ and $\{\alpha\gamma\epsilon\beta\delta\}$ will be changed into $-\{\alpha\beta\gamma\delta\epsilon\}$ and $-\{\alpha\gamma\epsilon\beta\delta\}$ respectively. Hence $\Pi_1$ remains unaltered by any one of the first five substitutions; and it is changed into $-\Pi_1$ by any one of the second five substitutions. And the like being the case as regards $\Pi_2$, &c., it follows that the quotient $\Pi_1 \div \Pi_2$, or say $P$, remains unaltered by any one of the ten substitutions. Now the 120 permutations of $\alpha$, $\beta$, $\gamma$, $\delta$, $\epsilon$ can be obtained as follows, viz. by forming the 12 different pentagons which can be formed with $\alpha$, $\beta$, $\gamma$, $\delta$, $\epsilon$ (treated as five points), and reading each of them off in either direction from any angle. To each of the 12 pentagons there corresponds a distinct value of $P$, but such value is not altered by the different modes of reading off the pentagon; $P$ is consequently a 12-valued function.

But there is a more simple form of the analytical expression of such a 12-valued function; in fact, if $[\alpha\beta\gamma\delta\epsilon]$ be any function which is not altered by any one of the above ten substitutions—if, for instance, $[\alpha\beta]$ is a symmetrical function of $x_\alpha$, $x_\beta$, and

$$[\alpha\beta\gamma\delta\epsilon] = [\alpha\beta] + [\beta\gamma] + [\gamma\delta] + [\delta\epsilon] + [\epsilon\alpha],$$

and therefore

$$[\alpha\gamma\epsilon\beta\delta] = [\alpha\gamma] + [\gamma\epsilon] + [\epsilon\beta] + [\beta\delta] + [\delta\alpha],$$

then any unsymmetrical function of $[\alpha\beta\gamma\delta\epsilon]$ and $[\alpha\gamma\epsilon\beta\delta]$ will be a 12-valued function homotypical with $P$.

Mr Jerrard's function $W$ is the sum of two values of his function $P$; the substitution by which the second is derived from the first can only be that which interchanges the two functions $[\alpha\beta\gamma\delta\epsilon]$ and $[\alpha\gamma\epsilon\beta\delta]$; and hence any symmetrical function of $[\alpha\beta\gamma\delta\epsilon]$ and $[\alpha\gamma\epsilon\beta\delta]$ is a function homotypical with Mr Jerrard's $W$; such symmetric function is in fact a 6-valued function only. Indeed it is easy to see that the twelve pentagons correspond together in pairs, either pentagon of a pair being derived from the other one by *stellation*, and the six values of the function in question corresponding to the six pairs of pentagons respectively.

Writing with Mr Cockle and Mr Harley,

$$\tau = x_\alpha x_\beta + x_\beta x_\gamma + x_\gamma x_\delta + x_\delta x_\epsilon + x_\epsilon x_\alpha,$$

$$\tau' = x_\alpha x_\gamma + x_\gamma x_\epsilon + x_\epsilon x_\beta + x_\beta x_\delta + x_\delta x_\alpha,$$

then $(\tau + \tau'$ is a symmetrical function of all the roots, and it must be excluded; but) $(\tau - \tau')^2$ or $\tau\tau'$ are each of them 6-valued functions of the form in question, and either of these functions is linearly connected with the Resolvent Product. In Lagrange's general theory of the solution of equations, if

$$f\iota = x_1 + \iota x_2 + \iota^2 x_3 + \iota^3 x_4 + \iota^4 x_5,$$

then the coefficients of the equation the roots whereof are $(f\iota)^5$, $(f\iota^2)^5$, $(f\iota^3)^5$, $(f\iota^4)^5$, and in particular the last coefficient $(f\iota f\iota^2 f\iota^3 f\iota^4)^5$, are determined by an equation of the sixth degree; and this last coefficient is a perfect fifth power, and its fifth root, or $f\iota f\iota^2 f\iota^3 f\iota^4$, is the function just referred to as the Resolvent Product.

The conclusion from the foregoing remarks is that *if the equation for $W$ has the above property of the rational expressibility of its roots*, the equation of the sixth order resulting from Lagrange's general theory has the same property.

I take the opportunity of adding a simple remark on cubic equations. The principle which furnishes what in a foregoing foot-note is called the *à priori* demonstration of Lagrange's theorem is that an equation need never contain extraneous roots; a quantity which has only one value will, if the investigation is properly conducted, be determined in the first instance by a linear equation; one which has two values by a quadratic equation, and so on; there is always enough, and not more than enough, to determine what is required.